Calculating Geometry

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Contents

Introduction

In this note, we shall try to solve some geometric problems in an algebraic way. This refers to finding the relationships between lengths of segments, angles and areas. The major tools that we need are trigonometric functions, since they connect side lengths and angles of triangles directly. Through expressing every length and angle we want in terms of some chosen references, a geometric problem can be reduced to a pure algebraic problem, which usually means to prove an identity.

Although doing geometric problems in such ways is usually tough, this might sometimes give a straightforward proof to the problem. By calculating some lengths and angles, we may sometimes obtain new information and deduce useful facts. This may also transform the problem to a simpler one which is easier to be done geometrically. Just like many other algebraic methods, knowledge in pure geometry could always help simplify our calculation. Therefore, learning pure geometry is still a must to solve geometric problems.

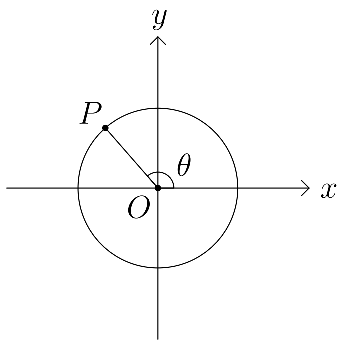
Section 1 mainly covers trigonometric functions. We shall introduce many identities which will be found useful later. In section 2, we see how these trigonometric functions are used for our purpose, mainly to solve some problems in triangle geometry. In section 3, we focus on some concurrency and collinearity problems. Lastly, in section 4, we introduce some more theorems of algebraic style and see some of their applications.

1 Trigonometric Functions

In this note, the main tools that we will use are the trigonometric functions. They are the simplest things that relate the two important ingredients in plane geometry, namely, angles and lengths. Having a thorough understanding of their properties is a must to go through the note.

We shall use the geometric definitions of the trigonometric functions for our purpose. Consider the coordinate plane with origin O. For each angle θ, let $P(a, b)$ be the point on the unit circle such that the directed angle measured anticlockwise from the positive x-axis to the ray OP is θ. Then we define three commonly used *trigonometric functions*(三角函數) as follows.

- *sine function*(正弦函數): $\sin\theta = b$
- *cosine function*(餘弦函數): $\cos\theta = a$
- *tangent function*(正切函數): $\tan\theta = \dfrac{b}{a}$ (which is only well-defined when $a \neq 0$)



There are other kinds of trigonometric functions, but we shall not discuss them in this note.

Proposition 1.1. Let θ be any angle measured in degree. Then the following hold.

(a) $\sin(\theta + 360°) = \sin\theta$

(b) $-1 \leqslant \sin\theta \leqslant 1$

(c) $0 < \sin\theta \leqslant 1$ for $0° < \theta < 180°$, increasing on $0° < \theta < 90°$ and decreasing on $90° < \theta < 180°$

(d) $\sin\theta = \sin(180° - \theta)$ and $\sin(-\theta) = -\sin\theta$

Also, the following table gives the values of $\sin\theta$ for some common angles θ.

θ	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0

As a corollary of parts (c) and (d), if x and y satisfy $0^\circ < x, y < 180^\circ$ and $\sin x = \sin y$, then we have $x = y$ or $x = 180^\circ - y$.

Proposition 1.2. Let θ be any angle measured in degree. Then the following hold.

(a) $\cos(\theta + 360^\circ) = \cos\theta$

(b) $-1 \leqslant \cos\theta \leqslant 1$

(c) $-1 < \cos\theta < 1$ and decreasing on $0^\circ < \theta < 180^\circ$

(d) $\cos\theta = -\cos(180^\circ - \theta)$ and $\cos(-\theta) = \cos\theta$

(e) $\cos\theta = \sin(90^\circ - \theta)$ and $\sin\theta = \cos(90^\circ - \theta)$

(f) $\sin^2\theta + \cos^2\theta = 1$

Also, the following table gives the values of $\cos\theta$ for some common angles θ.

θ	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1

Parts (e) and (f) provide some simple relations between the sine function and the cosine function. Note that (f) follows easily from Pythagoras' theorem.

Proposition 1.3. Let θ be any angle measured in degree. Then the following hold.

(a) $\tan(\theta + 180^\circ) = \tan\theta$

(b) $\tan 90^\circ$ is undefined

(c) $\tan\theta$ increases from 0 to ∞ on $0^\circ < \theta < 90^\circ$, and increases from $-\infty$ to 0 on $90^\circ < \theta < 180^\circ$

(d) $\tan\theta = -\tan(180^\circ - \theta)$ and $\tan(-\theta) = -\tan\theta$

(e) $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

Also, the following table gives the values of $\tan\theta$ for some common angles θ.

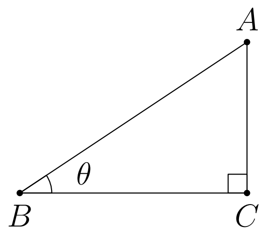
θ	0°	30°	45°	60°	120°	135°	150°	180°
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

Unlike sine and cosine, the tangent function is unbounded. In view of (e), we can express $\tan\theta$ as the sine and the cosine functions. Therefore, this function is only defined for convenience. In many applications of trigonometry in solving geometry problems, we mainly use the sine and the cosine functions.

Consider a right-angled $\triangle ABC$ with $\angle C = 90°$ and $\angle B = \theta$. By definition and by similar triangles, we have

$$\sin\theta = \frac{AC}{AB}, \quad \cos\theta = \frac{BC}{AB}, \quad \tan\theta = \frac{AC}{BC}.$$

This is the basic linkage between trigonometric functions and geometry.



We now give a number of algebraic relations between the trigonometric functions.

Proposition 1.4. (Compound angle formulae(和差公式)) Let x and y be any angles. Then the following hold.

(a) $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$

(b) $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$

(c) $\tan(x \pm y) = \dfrac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$

Since x and y can be angles of any size, the geometric proof of this proposition consists of various cases. Usually, most textbooks only deal with the usual case when x and y are acute. Anyway, we are not going to provide the detailed proof here since we are only interested in the application. By setting $x = y$, we have the following.

Proposition 1.5. (Double angle formulae(倍角公式)) Let x be any angle. Then the following hold.

(a) $\sin 2x = 2\sin x \cos x$

(b) $\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$

(c) $\tan 2x = \dfrac{2\tan x}{1 - \tan^2 x}$

Proposition 1.6. (Half angle formulae(半角公式)) Let x be any angle. Then the following hold.

(a) $\sin^2 \dfrac{x}{2} = \dfrac{1 - \cos x}{2}$

(b) $\cos^2 \dfrac{x}{2} = \dfrac{1 + \cos x}{2}$

This is just an alternative way to express the results in proposition 1.5.

Proposition 1.7. (Product-to-sum formulae(積化和差公式)) Let x and y be any angles. Then the following hold.

(a) $\sin x \cos y = \dfrac{1}{2}\big(\sin(x+y) + \sin(x-y)\big)$

(b) $\cos x \sin y = \dfrac{1}{2}\big(\sin(x+y) - \sin(x-y)\big)$

(c) $\cos x \cos y = \dfrac{1}{2}\big(\cos(x+y) + \cos(x-y)\big)$

(d) $\sin x \sin y = -\dfrac{1}{2}\big(\cos(x+y) - \cos(x-y)\big)$

These can be obtained from the compound angle formulae. For example, we have

$$\begin{aligned}\frac{1}{2}\big(\sin(x+y) + \sin(x-y)\big) &= \frac{1}{2}(\sin x \cos y + \cos x \sin y + \sin x \cos y - \cos x \sin y) \\ &= \sin x \cos y.\end{aligned}$$

Proposition 1.8. (Sum-to-product formulae(和差化積公式)) Let x and y be any angles. Then the following hold.

(a) $\sin x + \sin y = 2 \sin \dfrac{x+y}{2} \cos \dfrac{x-y}{2}$

(b) $\sin x - \sin y = 2 \cos \dfrac{x+y}{2} \sin \dfrac{x-y}{2}$

(c) $\cos x + \cos y = 2 \cos \dfrac{x+y}{2} \cos \dfrac{x-y}{2}$

(d) $\cos x - \cos y = -2 \sin \dfrac{x+y}{2} \sin \dfrac{x-y}{2}$

These follow from proposition 1.7 by changes of variables. For example, replacing x and y in

$$\sin x \cos y = \frac{1}{2}\big(\sin(x+y) + \sin(x-y)\big)$$

by $\dfrac{x+y}{2}$ and $\dfrac{x-y}{2}$ respectively, we obtain

$$\sin\frac{x+y}{2}\cos\frac{x-y}{2} = \frac{1}{2}(\sin x + \sin y).$$

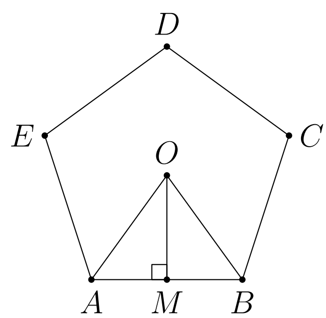
Since many of the above formulae look similar, it is not easy to memorize all of them. In fact, one may only memorize the compound angle formulae since the remaining ones can be deduced from these.

We first go through some simple problems to see how to use the trigonometric functions.

Problem 1.1. Let $ABCDE$ be a regular pentagon inscribed in the unit circle. Find its side length.

Analysis. Let O be the centre of the unit circle. Then $\triangle OAB$ is an isosceles triangle with $\angle AOB = 72°$ and $\angle OAB = \angle OBA = 54°$. Therefore, this problem is simply asking one to find $\cos 54° = \sin 36°$. One standard way to find the values of trigonometric functions involving multiples of $18°$ is to start from the relation $\sin 72° = \sin 108°$.

Solution.



Let O be the centre of the unit circle. Then $\angle AOB = \dfrac{360°}{5} = 72°$. Let M be the midpoint of AB. Since $OA = OB$, we have $\angle AOM = \dfrac{1}{2}\angle AOB = 36°$ and $AB = 2AM = 2\sin 36°$.

For convenience, let $\theta = 36°$. Noting $2\theta + 3\theta = 180°$, we have $\sin 2\theta = \sin 3\theta$. We apply the compound angle formula for the sine function and the double angle formulae as follows.

$$\begin{aligned}
& & \sin 2\theta &= \sin 3\theta \\
&\Rightarrow & 2\sin\theta\cos\theta &= \sin\theta\cos 2\theta + \cos\theta\sin 2\theta \\
&\Rightarrow & 2\sin\theta\cos\theta &= \sin\theta(2\cos^2\theta - 1) + \cos\theta(2\sin\theta\cos\theta) \\
&\Rightarrow & 2\cos\theta &= 4\cos^2\theta - 1 \\
&\Rightarrow & 4\cos^2\theta - 2\cos\theta - 1 &= 0
\end{aligned}$$

Viewing $\cos\theta$ as a variable, we easily get $\cos\theta = \dfrac{1 \pm \sqrt{5}}{4}$. Since θ is acute, the negative sign is rejected, and hence $\cos 36^\circ = \dfrac{1+\sqrt{5}}{4}$. Then

$$AB = 2\sin 36^\circ = 2\sqrt{1-\cos^2 36^\circ} = 2\sqrt{1 - \frac{6+2\sqrt{5}}{16}} = \frac{1}{2}\sqrt{10-2\sqrt{5}}.$$

□

Problem 1.2. In a rectangle $ABCD$, $AB = 1$ and $BC = 3$. The points E and F trisect the side AD, where E lies between A and F. Prove that

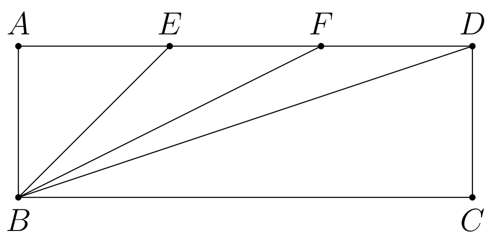
$$\angle DBC + \angle FBC = \angle EBC.$$

Analysis. For any real number x, we define $\tan^{-1} x$ to be the unique angle θ in $(-90^\circ, 90^\circ)$ such that $\tan\theta = x$. Then the desired equation is the same as

$$\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{2} = \tan^{-1} 1.$$

We can use the compound angle formula for the tangent function. The deduction is quite straightforward.

Solution.



Note that $AE = EF = FD = 1$. This implies

$$\begin{aligned}
\tan\angle DBC &= \frac{1}{3}, \\
\tan\angle FBC = \tan\angle AFB &= \frac{1}{2}, \\
\tan\angle EBC = \tan\angle AEB &= 1.
\end{aligned}$$

By the compound angle formula, we have

$$\tan(\angle DBC + \angle FBC) = \frac{\tan\angle DBC + \tan\angle FBC}{1 - \tan\angle DBC \tan\angle FBC} = \frac{\frac{1}{3} + \frac{1}{2}}{1 - \frac{1}{3} \times \frac{1}{2}} = 1 = \tan\angle EBC.$$

Since both $\angle DBC + \angle FBC$ and $\angle EBC$ lie in the interval $[0^\circ, 180^\circ)$, we must have $\angle DBC + \angle FBC = \angle EBC$. □

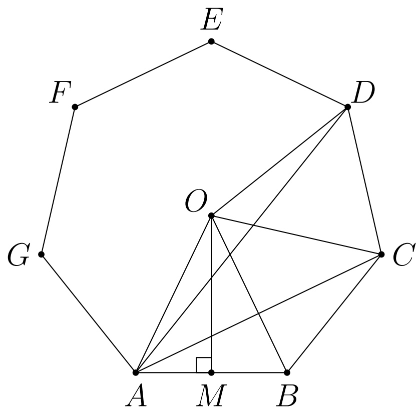
Remarks. Suppose we are going to show that two angles x and y are equal. By using trigonometric functions, we often obtain relations like $\sin x = \sin y$, $\cos x = \cos y$, $\tan x = \tan y$, etc. We should be careful in arguing that this implies $x = y$. This can usually be done by considering the sizes of the angles. Sometimes we may need to reject some cases using the configuration to conclude this.

Problem 1.3. Let $ABCDEFG$ be a regular heptagon. Prove that

$$\frac{1}{AB} = \frac{1}{AC} + \frac{1}{AD}.$$

Analysis. After rephrasing the problem in terms of the sine function, we need to prove an identity. This is a practice of using the product-to-sum formula and the sum-to-product formula. They are useful in proving identities in sine and cosine.

Solution.



Let O be the centre of the regular heptagon and let $r = OA$. Let M be the midpoint of AB. Since $\angle AOB = \frac{360^\circ}{7}$, we have $\angle AOM = \frac{180^\circ}{7}$. This implies

$$AB = 2AM = 2r\sin\frac{180^\circ}{7}.$$

Similarly, we find that

$$AC = 2r\sin\frac{360^\circ}{7}, \quad AD = 2r\sin\frac{540^\circ}{7}.$$

It remains to prove

$$\frac{1}{\sin\dfrac{180^\circ}{7}} = \frac{1}{\sin\dfrac{360^\circ}{7}} + \frac{1}{\sin\dfrac{540^\circ}{7}}.$$

Let $\theta = \dfrac{180^\circ}{7}$. The identity is equivalent to

$$\sin 2\theta \sin 3\theta = \sin\theta \sin 3\theta + \sin\theta \sin 2\theta.$$

Using the product-to-sum formula, we have

$$\begin{aligned}
& \sin 2\theta \sin 3\theta = \sin\theta \sin 3\theta + \sin\theta \sin 2\theta \\
\Leftrightarrow \quad & -\frac{1}{2}\big(\cos 5\theta - \cos(-\theta)\big) = -\frac{1}{2}\big(\cos 4\theta - \cos(-2\theta)\big) - \frac{1}{2}\big(\cos 3\theta - \cos(-\theta)\big) \\
\Leftrightarrow \quad & \cos 5\theta - \cos\theta = \cos 4\theta - \cos 2\theta + \cos 3\theta - \cos\theta \\
\Leftrightarrow \quad & \cos 5\theta + \cos 2\theta = \cos 4\theta + \cos 3\theta.
\end{aligned}$$

Next, we use the sum-to-product formula. This gives

$$\begin{aligned}
& \cos 5\theta + \cos 2\theta = \cos 4\theta + \cos 3\theta \\
\Leftrightarrow \quad & 2\cos\frac{7\theta}{2}\cos\frac{3\theta}{2} = 2\cos\frac{7\theta}{2}\cos\frac{\theta}{2}.
\end{aligned}$$

This is true since $\cos\dfrac{7\theta}{2} = \cos 90^\circ = 0$, so that both sides equal 0. Hence, the identity is established. $\square$

2 Triangle Geometry

Triangle geometry is the main kind of geometric problems in Olympiad. In many cases, the given figure is uniquely determined up to a ratio. Hence, we can express the length of every segment and every angle in terms of those of the given triangle theoretically. In this way, we are usually left to prove an identity in the last step. This sounds to be a reliable method to solve such geometric problems.

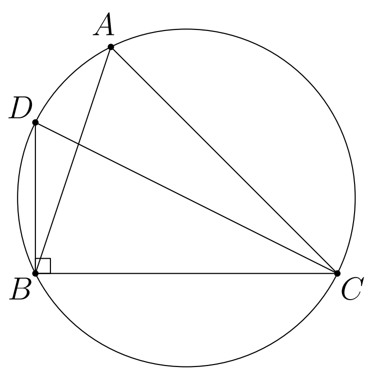
In the following, we use $[P_1P_2\cdots P_n]$ to denote the area of a polygon $P_1P_2\cdots P_n$. For a given $\triangle ABC$, we denote the following.

- a, b, c: the side lengths BC, CA, AB respectively
- R: the circumradius of $\triangle ABC$
- r: the inradius of $\triangle ABC$
- s: the semiperimeter $\dfrac{a+b+c}{2}$

In fact, we do not limit ourselves to triangles only. Knowledge in triangle geometry can be applied to solve some problems by finding the relationship between the lengths, angles and areas of triangles. The first major results that we need are the sine law and the cosine formula.

Theorem 2.1. (Sine law(正弦定理)) Let $\triangle ABC$ be any triangle. Then

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$



Proof. Let D be the point on (ABC) (which means the circumcircle of $\triangle ABC$) such that CD is a diameter. Then

$$\frac{a}{\sin A} = \frac{BC}{\sin \angle BDC} = CD = 2R$$

since $\triangle BCD$ is right-angled with $\angle DBC = 90°$. Similarly, we have the other two relations. □

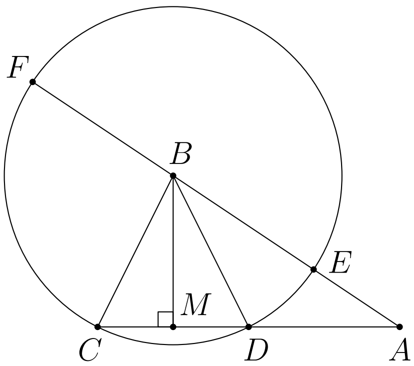
Sometimes the sine law refers to the result without the equality with $2R$, and the equality $\dfrac{a}{\sin A} = 2R$ is known as the extended sine law.

Theorem 2.2. (Cosine formula(餘弦定理)) Let $\triangle ABC$ be any triangle. Then

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

Equivalently,

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$



Proof. Let M be the projection of B on the line AC. Suppose the circle with centre B passing through C cuts the line AB at E and F, and cuts the line AC again at D. Note that $a\cos C = CM$ where directed length is used with the choice that CA is positive. Since M is the midpoint of CD, we have $2a\cos C = CD$, and hence $b - 2a\cos C = CA - CD = DA$. As C, D, E, F are concyclic, we have

$$\begin{aligned} b^2 - 2ab\cos C &= b(b - 2a\cos C) = CA \times DA = EA \times FA \\ &= (BA - BE)(BA + FB) = (c-a)(c+a) = c^2 - a^2. \end{aligned}$$

The result then follows. $\square$

As a special case, we have the following well-known theorem.

Theorem 2.3. (Pythagoras' theorem(畢氏定理)) Let $\triangle ABC$ be any triangle. Then $C = 90^\circ$ if and only if(iff) $c^2 = a^2 + b^2$.

The sine law and the cosine formula provide the simplest relations between the angles and the side lengths of a triangle. By knowing any three of $a, b, c, \angle A, \angle B, \angle C$, the remaining ones can be found using these two theorems (in terms of sine and cosine).

Many pure geometry problems consist of similar triangles. They provide a lot of relations between angles and lengths to us. In case one cannot observe any pair of similar triangles, one can still find the corresponding relationship by applying the sine law and/or the cosine formula.

For example, in $\triangle ABC$ and $\triangle XYZ$, suppose we know $\angle A = \angle X$ and $\angle B = \angle Y$. Normally, we can use these to prove $\triangle ABC \sim \triangle XYZ$, and then deduce a result $\frac{BC}{AC} = \frac{YZ}{XZ}$ which may be useful in solving the problem. If, for some unknown reasons, we cannot spot this pair of similar triangles, we can still use the sine law to deduce

$$\frac{BC}{AC} = \frac{\sin A}{\sin B} = \frac{\sin X}{\sin Y} = \frac{YZ}{XZ}.$$

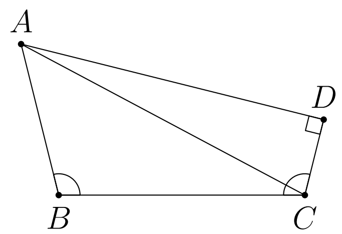
So this can be viewed as a less clever but more straightforward way to obtain a solution.

By using these two basic theorems alone (in particular, the sine law), we can already solve many Olympiad problems.

Problem 2.1. (Omaforos Contest 2020 Problem 7) Let $ABCD$ be a convex quadrilateral in which $\angle ABC = \angle BCD > 90°$ and $\angle CDA = 90°$. In this quadrilateral, it is also true that $AB = 2CD$. Prove that the bisector of $\angle ACB$ is perpendicular to CD.

Analysis. The setting of this problem is simple, and the only conditions are lengths and angles. Therefore, there is a high chance that we can use trigonometric functions to solve the problem efficiently. We usually first think of the sine law because this result is much simpler. On the other hand, the cosine formula involves the square of lengths, which makes things complicated.

Solution.



Let $x = \angle ACB$ and $y = \angle DCA$. In $\triangle ABC$, since we are given some conditions involving $\angle CBA$ and AB, we should apply the sine law to obtain

$$\frac{AB}{AC} = \frac{\sin \angle ACB}{\sin \angle CBA} = \frac{\sin x}{\sin (x+y)}. \tag{1}$$

In $\triangle ACD$, we easily obtain

$$\frac{CD}{AC} = \cos \angle DCA = \cos y. \tag{2}$$

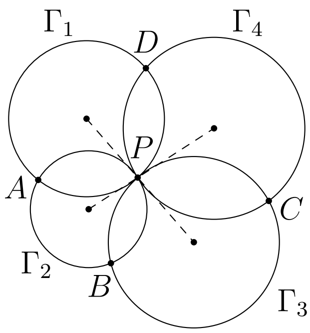
Combining (1) and (2), and using $AB = 2CD$, we find that $\dfrac{\sin x}{\sin(x+y)} = 2\cos y$. Now, by the product-to-sum formula, we have

$$\begin{aligned} & \frac{\sin x}{\sin(x+y)} = 2\cos y \\ \Rightarrow \quad & \sin x = 2\sin(x+y)\cos y \\ \Rightarrow \quad & \sin x = \sin(x+2y) + \sin x \\ \Rightarrow \quad & \sin(x+2y) = 0. \end{aligned}$$

As $x + 2y > 0°$ and $x + 2y < 2(x + y) < 360°$, we must have $x + 2y = 180°$. This implies $\frac{x}{2} + y = 90°$, which means exactly that the bisector of $\angle ACB$ is perpendicular to the line CD. $\square$

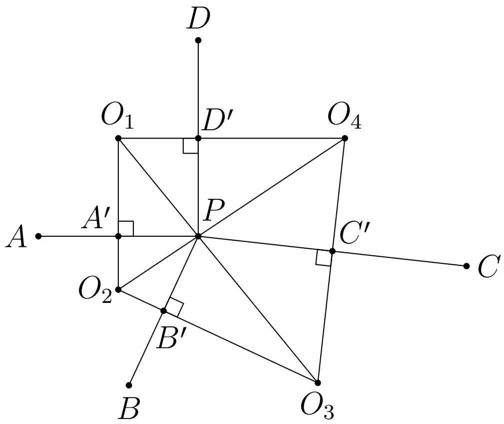
Problem 2.2. (IMO Shortlist 2003 G4) Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be distinct circles such that Γ_1, Γ_3 are externally tangent at P, and Γ_2, Γ_4 are externally tangent at the same point P. Suppose that Γ_1 and Γ_2; Γ_2 and Γ_3; Γ_3 and Γ_4; Γ_4 and Γ_1 meet at A, B, C, D, respectively, and that all these points are different from P. Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}.$$



Analysis. We are given several circles some of which are tangent to each other. Usually, we try to locate the centres of the circles to use the tangential property. Let O_j be the centre of each Γ_j. Then P is the intersection of O_1O_3 and O_2O_4. Also, since A and P are the intersection points of Γ_1 and Γ_2, we know that they are symmetric in the line O_1O_2. These properties are indeed equivalent to what we have. Indeed, we may reconstruct the figure starting from a convex quadrilateral $O_1O_2O_3O_4$ and define points P, A, B, C, D using the facts we have found. The advantage is that we need not consider the circles.

Solution.



Let O_1, O_2, O_3, O_4 be the centres of $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ respectively. Since Γ_1, Γ_3 are tangent at P, points O_1, P, O_3 are collinear. Similarly, O_2, P, O_4 are collinear. Therefore, P is the intersection of O_1O_3 and O_2O_4.

Note that O_1O_2 is the perpendicular bisector of AP. Therefore, the midpoint A' of AP is the projection from P on the line O_1O_2. Similarly, define B', C', D' to be the midpoints of BP, CP, DP respectively.

After reconstructing the figure, we start our proof using trigonometry. One use of introducing the points A', B', C', D' is that we get some concyclic points. For example, O_1, A', P, D' all lie on the circle with diameter O_1P. This allows us to relate the angles formed by 3 of these 4 points using the sine law. Firstly, we note that the conclusion is equivalent to

$$\frac{A'B' \cdot B'C'}{A'D' \cdot D'C'} = \frac{PB'^2}{PD'^2}$$

since we have, for example, $AB = 2A'B'$ by the midpoint theorem and $PB = 2PB'$, etc. We rewrite this as

$$\frac{A'B'}{PB'} \cdot \frac{B'C'}{PB'} \cdot \frac{PD'}{A'D'} \cdot \frac{PD'}{D'C'} = 1. \tag{1}$$

The reason for writing in this way is that the lengths in each fraction are comparable using the sine law, as they are sides of a certain triangle. We have

$$\frac{A'B'}{PB'} = \frac{\sin \angle A'PB'}{\sin \angle PA'B'} = \frac{\sin \angle A'O_2B'}{\sin \angle PO_2B'} = \frac{\sin \angle O_1O_2O_3}{\sin \angle O_4O_2O_3}.$$

The second equality holds since P, A', O_2, B' are concyclic. Indeed, if one is familiar with the sine law, one can skip the first equality and get the same result immediately since $\angle A'O_2B'$ is an angle opposite to the chord $A'B'$ on the circle, etc.

The other fractions in (1) can be transformed in the same way. We can show that

(1) is equivalent to

$$\frac{\sin \angle O_1O_2O_3}{\sin \angle O_4O_2O_3} \cdot \frac{\sin \angle O_2O_3O_4}{\sin \angle O_1O_3O_2} \cdot \frac{\sin \angle O_3O_1O_4}{\sin \angle O_4O_1O_2} \cdot \frac{\sin \angle O_2O_4O_1}{\sin \angle O_3O_4O_1} = 1. \tag{2}$$

Note that this equation to be shown only consists of points O_1, O_2, O_3, O_4. It has nothing to do with the remaining points and hence should be true for any quadrilateral $O_1O_2O_3O_4$. Hence, it is indeed a simple identity. To prove (2), we again rearrange the terms so that each term can be simplified using the sine law. We find that (2) is the same as

$$\frac{\sin \angle O_1O_2O_3}{\sin \angle O_1O_3O_2} \cdot \frac{\sin \angle O_2O_3O_4}{\sin \angle O_4O_2O_3} \cdot \frac{\sin \angle O_3O_1O_4}{\sin \angle O_3O_4O_1} \cdot \frac{\sin \angle O_2O_4O_1}{\sin \angle O_4O_1O_2} = 1.$$

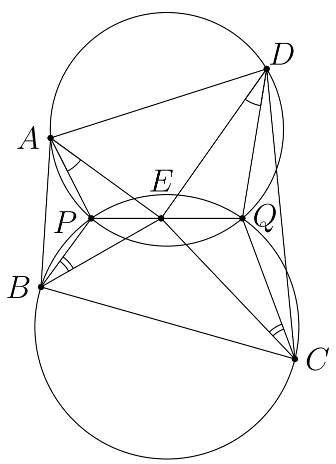
Applying the sine law to each fraction, this means we have to prove

$$\frac{O_1O_3}{O_1O_2} \cdot \frac{O_2O_4}{O_3O_4} \cdot \frac{O_3O_4}{O_1O_3} \cdot \frac{O_1O_2}{O_2O_4} = 1.$$

This is obviously true. □

Remarks. In this solution, we have rearranged the terms many times. The aim is to put related things together so that we can simplify them using the sine law. Therefore, the steps are rather natural.

Problem 2.3. (IMO Shortlist 2008 G3) Let $ABCD$ be a convex quadrilateral and let P and Q be points in $ABCD$ such that $PQDA$ and $QPBC$ are cyclic quadrilaterals. Suppose that there exists a point E on the line segment PQ such that $\angle PAE = \angle QDE$ and $\angle PBE = \angle QCE$. Show that the quadrilateral $ABCD$ is cyclic.

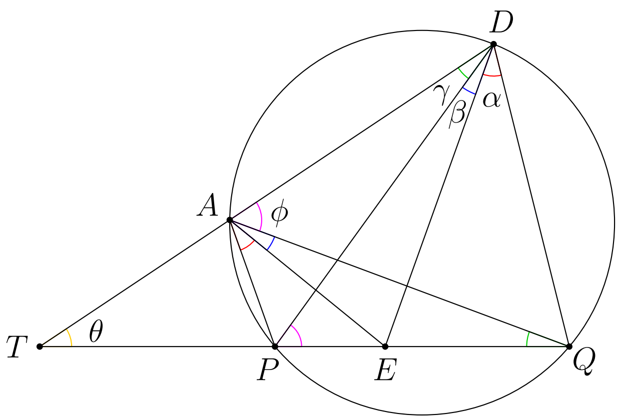


Analysis. Since the problem only involves cyclic quadrilaterals and some equal angles, the sine law could be useful if the correct triangles are chosen. To show that

$ABCD$ is cyclic, it seems that using angle chasing alone is not sufficient as the angles are not so related to each other. We shall prove it using power of points. This means we need some side length calculations, and hence suggesting the use of the sine law again.

Since the assertion must be true, the lines DA, QP, CB are either concurrent or parallel. As the configuration is symmetric in the line PQ, we may simply focus on one part of it. In fact, we shall prove that the location of the intersection point of DA and QP (if exists) depends only on P, E, Q. Then by symmetry, this is also the intersection point of CB and QP. This is an extremely useful technique when dealing with a symmetric figure.

Solution.



Firstly, suppose DA intersects QP at some point T. Denote $\angle PAE = \angle QDE = \alpha$,

$$\angle QAE = \angle QAP - \angle PAE = \angle QDP - \angle QDE = \angle PDE = \beta,$$

$\angle ADP = \angle AQP = \gamma$, $\angle DAQ = \angle DPQ = \phi$ and $\angle DTQ = \theta$.

As discussed, we would like to find the lengths of TP and TQ. Therefore, we find triangles including them as one of the sides. In $\triangle TPD$, we get

$$\frac{TP}{\sin\gamma} = \frac{DP}{\sin\theta} = \frac{TD}{\sin(180^\circ - \phi)} = \frac{TD}{\sin\phi}. \tag{1}$$

In $\triangle TQD$, we get

$$\frac{TQ}{\sin(\alpha+\beta+\gamma)} = \frac{DQ}{\sin\theta} = \frac{TD}{\sin(180^\circ - \alpha - \beta - \phi)} = \frac{TD}{\sin(\alpha+\beta+\phi)}. \tag{2}$$

As we want to convert these to the lengths PE and QE, we consider triangles with these sides. Applying the sine law in $\triangle PDE$, $\triangle PAE$, $\triangle QAE$ and $\triangle QDE$

respectively, we get

$$\frac{PE}{\sin\beta} = \frac{DE}{\sin\phi}, \tag{3}$$
$$\frac{PE}{\sin\alpha} = \frac{AE}{\sin(180^\circ - \alpha - \beta - \gamma)} = \frac{AE}{\sin(\alpha + \beta + \gamma)}, \tag{4}$$
$$\frac{QE}{\sin\beta} = \frac{AE}{\sin\gamma}, \tag{5}$$
$$\frac{QE}{\sin\alpha} = \frac{DE}{\sin(180^\circ - \alpha - \beta - \phi)} = \frac{DE}{\sin(\alpha + \beta + \phi)}. \tag{6}$$

Then we are able to use these to clear all angles and get our desired result, i.e. a relationship between TP, TQ, PE, QE. We have

$$\frac{TP}{TD} \overset{(1)}{=} \frac{\sin\gamma}{\sin\phi} \overset{(3),(5)}{=} \frac{\frac{AE}{QE}\sin\beta}{\frac{DE}{PE}\sin\beta} = \frac{AE}{DE}\cdot\frac{PE}{QE}$$

and

$$\frac{TQ}{TD} \overset{(2)}{=} \frac{\sin(\alpha+\beta+\gamma)}{\sin(\alpha+\beta+\phi)} \overset{(4),(6)}{=} \frac{\frac{AE}{PE}\sin\alpha}{\frac{DE}{QE}\sin\alpha} = \frac{AE}{DE}\cdot\frac{QE}{PE}.$$

These yield

$$\frac{TP}{TQ} = \frac{\frac{TP}{TD}}{\frac{TQ}{TD}} = \frac{\frac{AE}{DE}\cdot\frac{PE}{QE}}{\frac{AE}{DE}\cdot\frac{QE}{PE}} = \left(\frac{PE}{QE}\right)^2.$$

If CB and QP intersect at a point T', then $\frac{T'P}{T'Q} = \left(\frac{PE}{QE}\right)^2 = \frac{TP}{TQ}$ by symmetry. As $PQDA$ and $QPBC$ are convex, both T, T' lie outside the segment QP. Therefore, we must have $T = T'$, and hence DA, QP, CB are concurrent. Then

$$TA\cdot TD = TP\cdot TQ = TB\cdot TC,$$

implying that $ABCD$ is a cyclic quadrilateral.

It remains to consider the case when at least one of AD, BC is parallel to PQ. Without loss of generality(WLOG), assume $BC//PQ$. Then $PBCQ$ is an isosceles trapezoid and E is the midpoint of PQ by symmetry. This implies AD and PQ are parallel as well, since otherwise, we get $\frac{TP}{TQ} = \left(\frac{PE}{QE}\right)^2 = 1$ as in the general case, which is impossible. Then $ABCD$ is an isosceles trapezoid by symmetry, and is therefore cyclic.

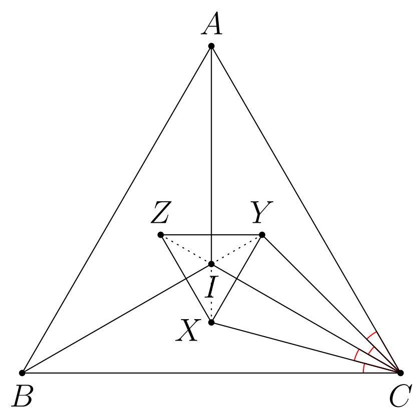
Therefore, $ABCD$ is cyclic in any case. □

Remarks. This solution suggests one helpful way when applying the sine law, i.e. labelling all equal angles in the figure. This helps us spot the required triangles easily. We only need to write down all relations obtained from the sine law, and combine them together to get the desired result.

Problem 2.4. (IMO Shortlist 2009 G7) Let ABC be a triangle with incentre I and let X, Y and Z be the incentres of the triangles BIC, CIA and AIB, respectively. Let the triangle XYZ be equilateral. Prove that ABC is equilateral too.

Analysis. The starting point of solving this problem using trigonometry is obvious since we have nothing apart from the equal angles and equal sides. The only trick of the following solution is to use the identities of trigonometric functions in a clever way.

Solution.



Let

$$\angle BAZ = \angle IAZ = \angle IAY = \angle CAY = \alpha,$$
$$\angle CBX = \angle IBX = \angle IBZ = \angle ABZ = \beta,$$
$$\angle ACY = \angle ICY = \angle ICX = \angle BCX = \gamma.$$

Then $\alpha + \beta + \gamma = 45°$. Note that the interior angles of $\triangle XYZ$ cannot be easily found in terms of α, β, γ. Therefore, to use the equilateral triangle condition, we shall work on the relation $XY = XZ$. To find the length XY, one way is to apply the cosine formula on $\triangle IXY$ since IX, IY and $\angle XIY$ are easy to find. Firstly, we apply the sine law to $\triangle ICX$ and $\triangle ICY$. These give

$$\frac{IC}{\sin(90° + \beta)} = \frac{IC}{\sin \angle IXC} = \frac{IX}{\sin \angle ICX} = \frac{IX}{\sin \gamma}$$

and

$$\frac{IC}{\sin(90° + \alpha)} = \frac{IC}{\sin \angle IYC} = \frac{IY}{\sin \angle ICY} = \frac{IY}{\sin \gamma}.$$

Comparing these two equations, we get

$$\frac{IX}{IY} = \frac{\sin(90^\circ + \alpha)}{\sin(90^\circ + \beta)} = \frac{\cos\alpha}{\cos\beta}.$$

Similarly, we have $IX : IY : IZ = \cos\alpha : \cos\beta : \cos\gamma$. As the problem does not depend on the size of the figure, we may assume $IX = \cos\alpha$, $IY = \cos\beta$ and $IZ = \cos\gamma$ for convenience.

Next, we find

$$\angle XIY = \angle CIX + \angle CIY = \frac{\angle BIC}{2} + \frac{\angle AIC}{2} = 45^\circ + \alpha + 45^\circ + \beta = 135^\circ - \gamma.$$

By symmetry, $\angle XIZ = 135^\circ - \beta$. Then we can apply the cosine formula as follows.

$$\begin{aligned}
& XY^2 = XZ^2 \\
\Rightarrow\quad & IX^2 + IY^2 - 2IX \cdot IY \cos\angle XIY = IX^2 + IZ^2 - 2IX \cdot IZ \cos\angle XIZ \\
\Rightarrow\quad & \cos^2\beta - 2\cos\alpha\cos\beta\cos(135^\circ - \gamma) = \cos^2\gamma - 2\cos\alpha\cos\gamma\cos(135^\circ - \beta) \\
\Rightarrow\quad & \cos^2\beta - \cos^2\gamma = 2\cos\alpha\big(\cos\beta\cos(135^\circ - \gamma) \\
& \qquad - \cos\gamma\cos(135^\circ - \beta)\big). \qquad (1)
\end{aligned}$$

To simplify the right-hand side, we use the product-to-sum formula. This gives

$$\begin{aligned}
& \cos\beta\cos(135^\circ - \gamma) - \cos\gamma\cos(135^\circ - \beta) \\
={}& \frac{1}{2}\big(\cos(135^\circ + \beta - \gamma) + \cos(\beta + \gamma - 135^\circ)\big) \\
& - \frac{1}{2}\big(\cos(135^\circ + \gamma - \beta) + \cos(\beta + \gamma - 135^\circ)\big) \\
={}& \frac{1}{2}\big(\cos(135^\circ + \beta - \gamma) - \cos(135^\circ + \gamma - \beta)\big).
\end{aligned}$$

This can be further simplified using the sum-to-product formula. Indeed,

$$\frac{1}{2}\big(\cos(135^\circ + \beta - \gamma) - \cos(135^\circ + \gamma - \beta)\big) = -\sin 135^\circ \sin(\beta - \gamma) = -\frac{1}{\sqrt{2}}\sin(\beta - \gamma).$$

Hence, (1) becomes

$$\cos^2\beta - \cos^2\gamma = -\sqrt{2}\cos\alpha\sin(\beta - \gamma). \qquad (2)$$

Again, we use the product-to-sum formula to get

$$\begin{aligned}
& -\sqrt{2}\cos\alpha\sin(\beta - \gamma) \\
={}& -\frac{1}{\sqrt{2}}\big(\sin(\alpha + \beta - \gamma) - \sin(\alpha - \beta + \gamma)\big) \\
={}& -\frac{1}{\sqrt{2}}\big(\sin(45^\circ - 2\gamma) - \sin(45^\circ - 2\beta)\big)
\end{aligned}$$

$$= -\frac{1}{\sqrt{2}}(\sin 45° \cos 2\gamma - \cos 45° \sin 2\gamma - \sin 45° \cos 2\beta + \cos 45° \sin 2\beta)$$
$$= -\frac{1}{2}(\cos 2\gamma - \sin 2\gamma - \cos 2\beta + \sin 2\beta).$$

As we get some trigonometric functions involving 2β and 2γ, it would be good to transform the left-hand side $\cos^2\beta - \cos^2\gamma$ to these angles as well. We have

$$\cos^2\beta - \cos^2\gamma = \frac{1+\cos 2\beta}{2} - \frac{1+\cos 2\gamma}{2}.$$

Therefore, after simplification, (2) implies $\sin 2\beta = \sin 2\gamma$. Then we get $\beta = \gamma$ or $2\beta + 2\gamma = 180°$. Since $\beta + \gamma < \alpha + \beta + \gamma = 45° < 90°$, the latter case is rejected. Hence, $\beta = \gamma$. By symmetry, we have $\alpha = \beta = \gamma$. This means $\triangle ABC$ is equilateral. $\square$

Remarks. The trick used in this solution is simple. We only keep on using the product-to-sum formulae, the sum-to-product formulae, etc. This can always simplify a symmetric expression as the symmetric terms cancel each other in most cases.

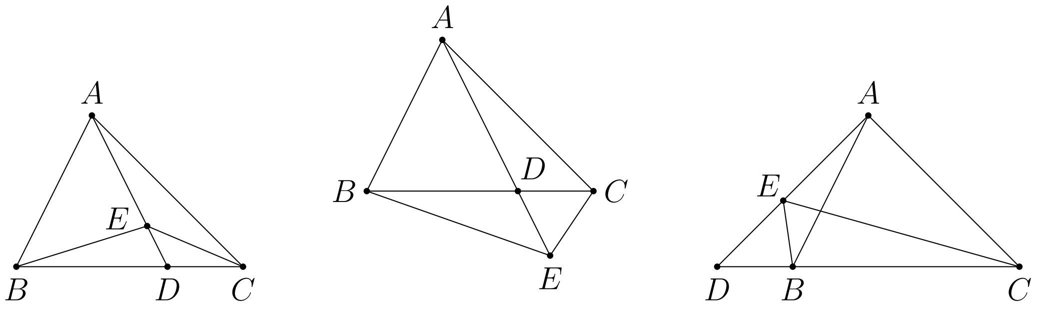
Apart from the sine law and the cosine formula, the area of a triangle is an useful ingredient that helps relate different quantities together. Firstly, we have the following result about ratio of areas.

Proposition 2.1. In $\triangle ABC$, let D be a point on the line BC, and let E be a point on the line AD. Then

$$\frac{[ABE]}{[ACE]} = \frac{BD}{CD}.$$

In particular, we have

$$\frac{[ABD]}{[ACD]} = \frac{BD}{CD}.$$



Proof. The above figures give a few possible configurations. But there are far a lot more different configurations in which the proposition is applicable. In this proof, we only provide a proof to the first configuration, i.e. when D lies on the segment BC and E lies on the segment AD.

In $\triangle ABD$ and $\triangle ACD$, they share the same height h from the vertex A. Therefore, we have

$$\frac{[ABD]}{[ACD]} = \frac{\frac{1}{2}BD \cdot h}{\frac{1}{2}CD \cdot h} = \frac{BD}{CD}.$$

This proves the result for the special case. Similarly, by considering $\triangle EBD$ and $\triangle ECD$, we have

$$\frac{[EBD]}{[ECD]} = \frac{BD}{CD}.$$

Combining these, we obtain

$$\frac{[ABE]}{[ACE]} = \frac{[ABD]-[EBD]}{[ACD]-[ECD]} = \frac{\frac{BD}{CD}[ACD] - \frac{BD}{CD}[ECD]}{[ACD]-[ECD]} = \frac{BD}{CD}.$$

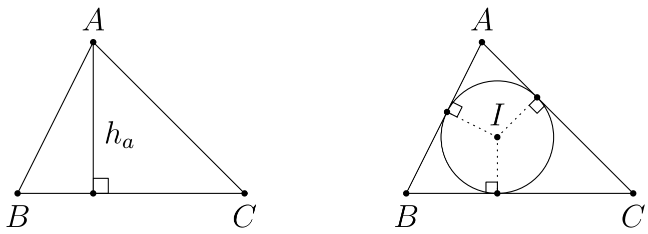
□

Another more important result about areas of triangles is given below.

Proposition 2.2. Let $\triangle ABC$ be any triangle. Then

$$\begin{aligned}[ABC] &= \frac{1}{2}ab\sin C = \frac{abc}{4R} = sr = \sqrt{s(s-a)(s-b)(s-c)} \\ &= \frac{1}{4}\sqrt{-a^4-b^4-c^4+2a^2b^2+2b^2c^2+2c^2a^2}.\end{aligned}$$

Proof. It is easy to see that $b\sin C$ equals the length of the height h_a from vertex A. Therefore, we have $[ABC] = \frac{1}{2}ah_a = \frac{1}{2}ab\sin C$. Using the sine law, this is equal to $\frac{abc}{4R}$.



Let I be the incentre of $\triangle ABC$. Then

$$[ABC] = [IBC] + [ICA] + [IAB] = \frac{ar}{2} + \frac{br}{2} + \frac{cr}{2} = sr.$$

Next, $[ABC] = \sqrt{s(s-a)(s-b)(s-c)}$ is called Heron's formula(海倫公式). By substituting $s = \frac{a+b+c}{2}$, it is identical to $\frac{1}{4}\sqrt{-a^4-b^4-c^4+2a^2b^2+2b^2c^2+2c^2a^2}$.

Therefore, we shall only prove the latter. In fact, by the cosine formula, we have

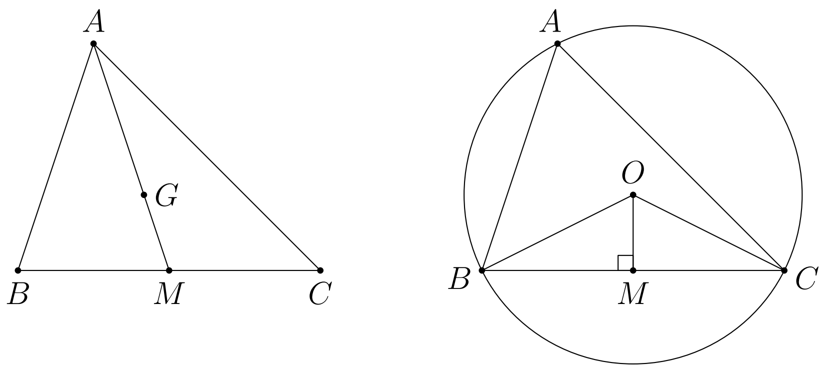
$$\begin{aligned}[ABC] &= \frac{1}{2}ab\sin C = \frac{1}{2}ab\sqrt{1-\cos^2 C} = \frac{1}{2}ab\sqrt{1-\frac{(a^2+b^2-c^2)^2}{4a^2b^2}} \\ &= \frac{1}{4}\sqrt{4a^2b^2-(a^2+b^2-c^2)^2} = \frac{1}{4}\sqrt{-a^4-b^4-c^4+2a^2b^2+2b^2c^2+2c^2a^2}\end{aligned}$$

as desired. □

In application, the use of proposition 2.2 is not to find the area of a triangle. Indeed, the formulae relate different important terms together including R, r and s. Sometimes it is easier to express some lengths in terms of these symbols, and we can convert them to a, b, c using the area formulae.

Before we see some more problems, let us revise some facts related to the common triangle centres. We only focus on acute triangles in the discussion below. One should be aware of some necessary modification in the general case.

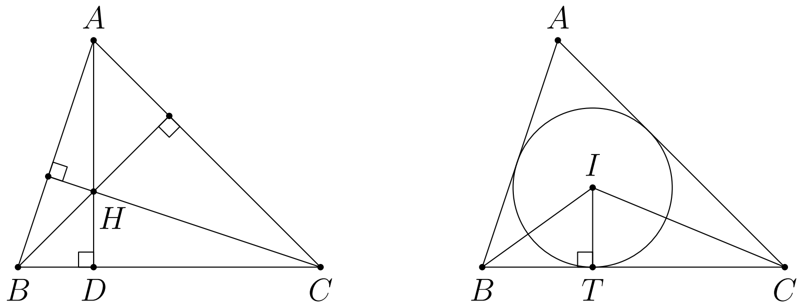
In $\triangle ABC$, let G be the centroid and let M be the midpoint of BC. Then we have $AG : GM = 2 : 1$ and $AM = \frac{1}{2}\sqrt{2b^2+2c^2-a^2}$. The ratio that the centroid cuts a median is the most useful property of the centroid, while the formula for the length of the median is called Apollonius' theorem(阿波羅尼斯定理). It can be proved by using the cosine formula in $\triangle ABM$.



Let O be the circumcentre of $\triangle ABC$. By definition, we have $OA = OB = OC = R$. There are some angles related to O which are commonly used, such as $\angle BOC = 2A$, $\angle BOM = A$ and $\angle OBC = 90° - A$. For example, this yields $OM = R\cos A$.

Let H be the orthocentre of $\triangle ABC$, and let D be the foot of altitude from A. The length AD can be expressed in terms of the area, i.e. $AD = \frac{2[ABC]}{a}$. Since there are a lot of right-angled triangles in a configuration with altitudes and the orthocentre, it is not hard to use trigonometric functions to relate the lengths of segments. For example, we have $BD = AB\cos B$ and $DH = BD\tan\angle DBH = \frac{AB\cos B}{\tan C}$. Another commonly used result is $AH = 2R\cos A$, since $AH = 2OM$ by $\triangle AGH \sim \triangle MGO$.

There are also many concyclic points in the figure, which leads to many groups of equal angles and supplementary angles. For example, we have $\angle BHC = 180° - A$.



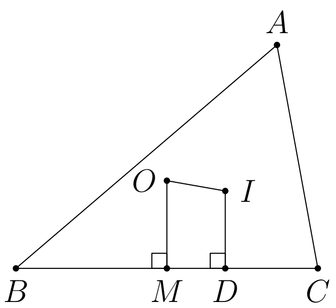
Let I be the incentre of $\triangle ABC$, and let T be the contact point of the incircle and the side BC. By definition, we have $IT = r$. Also, it can be found that $BT = s - b$ and $CT = s - c$. Sometimes it is simpler to express everything in terms of $s - a$, $s - b$ and $s - c$ instead of a, b, c, since the change of variables is reversible and the condition that a, b, c satisfying the triangle inequality is the same as $s - a, s - b, s - c > 0$. Some other useful results include $\angle BIC = 90° + \dfrac{A}{2}$ and $BI = \dfrac{r}{\sin\frac{B}{2}}$, etc.

There are analogous results for excentres since the incentre and the excentres share many similar properties. In addition to these properties, some elementary results like the internal and the external angle bisector theorem, and the power of a point theorem are quite helpful when working on some length chasing.

Problem 2.5. (Euler's formula(歐拉公式)) Let O and I be the circumcentre and the incentre of $\triangle ABC$ respectively. Prove that $OI^2 = R^2 - 2Rr$.

Analysis. The line OI is not related to the sides of $\triangle ABC$ directly. A useful trick is to draw perpendicular lines from O and I to the sides. This helps construct some right-angled triangles, so that we can apply Pythagoras' theorem. This makes the calculation simpler since there is no need to use the sine law or the cosine formula.

Solution.



Let M and D be the feet of perpendiculars from O and I to the side BC respectively. Note that M is the midpoint of BC. WLOG assume $BM \leqslant BD$ (which means $b \leqslant c$).

We have $OM = R\cos A$, $ID = r$ and

$$MD = BD - BM = (s-b) - \frac{a}{2} = \frac{c-b}{2}.$$

By Pythagoras' theorem, we obtain

$$\begin{aligned}
OI^2 = (OM - ID)^2 + MD^2 &= (R\cos A - r)^2 + \left(\frac{c-b}{2}\right)^2 \\
&= R^2\cos^2 A - 2Rr\cos A + r^2 + \left(\frac{c-b}{2}\right)^2 \\
&= R^2(1-\sin^2 A) - 2Rr\cdot\frac{b^2+c^2-a^2}{2bc} + r^2 + \left(\frac{c-b}{2}\right)^2.
\end{aligned}$$

Here, we apply the identity $\cos^2 A = 1 - \sin^2 A$ to extract the term R^2 since there is such a term in the final result. For the remaining term involving R, we use the sine law $R\sin A = \frac{a}{2}$. Similarly, we can extract the term $2Rr$ from the second term artificially. This gives

$$\begin{aligned}
&OI^2 \\
&= R^2 - \frac{a^2}{4} - 2Rr - Rr\cdot\frac{b^2-2bc+c^2-a^2}{bc} + r^2 + \left(\frac{c-b}{2}\right)^2 \\
&= R^2 - 2Rr - \frac{a^2}{4} - \frac{abc}{4[ABC]}\cdot\frac{[ABC]}{s}\cdot\frac{b^2-2bc+c^2-a^2}{bc} + \left(\frac{[ABC]}{s}\right)^2 + \left(\frac{c-b}{2}\right)^2 \\
&= R^2 - 2Rr + \frac{1}{4}\left((c-b)^2 - a^2\right) - \frac{a\left((b-c)^2-a^2\right)}{4s} + \frac{[ABC]^2}{s^2} \\
&= R^2 - 2Rr + \frac{(c-b-a)(c-b+a)}{4} - \frac{a(b-c-a)(b-c+a)}{4s} + \frac{s(s-a)(s-b)(s-c)}{s^2} \\
&= R^2 - 2Rr - (s-b)(s-c) + \frac{a(s-b)(s-c)}{s} + \frac{(s-a)(s-b)(s-c)}{s} \\
&= R^2 - 2Rr + (s-b)(s-c)\left(-1 + \frac{a}{s} + \frac{s-a}{s}\right) \\
&= R^2 - 2Rr.
\end{aligned}$$

□

Remarks. There are many ways to prove the identity even in an algebraic way. In this solution, we mainly use two techniques. Firstly, we extract the term $R^2 - 2Rr$ from the expression at an early stage so that the remaining terms should sum up to zero. This can always give simpler calculations since zero has infinitely many factors for which we can take out.

Secondly, we use different area formulae given in proposition 2.2 to relate the terms $R, r, [ABC]$ to a, b, c. As a, b, c are independent in this problem, the remaining expression must be an identity.

Note that $1 \pm \cos A = \dfrac{2bc \pm (b^2 + c^2 - a^2)}{2bc}$ can be factorized as shown in the solution. This factorization is often used in solutions involving the cosine formula.

Next, we use Euler's formula to solve the following problem.

Problem 2.6. Let O and I be the circumcentre and the incentre of $\triangle ABC$ respectively. Prove that $\angle AIO = 90^\circ$ iff $b + c = 2a$.

Analysis. To handle the right angle, we shall apply Pythagoras' theorem to get an equivalent statement to $\angle AIO = 90^\circ$. The lengths AI and AO can be easily obtained while OI can be obtained via Euler's formula. Then we use the same trick as in problem 2.5 to convert everything to a, b, c.

Solution. Note that $AO = R$, $AI = \dfrac{r}{\sin \frac{A}{2}}$ and $OI = \sqrt{R^2 - 2Rr}$ by Euler's formula. We have

$$\begin{aligned}
& \angle AIO = 90^\circ \\
\Leftrightarrow \quad & AI^2 + OI^2 = AO^2 \\
\Leftrightarrow \quad & \frac{r^2}{\sin^2 \frac{A}{2}} + R^2 - 2Rr = R^2 \\
\Leftrightarrow \quad & r = 2R \sin^2 \frac{A}{2} \\
\Leftrightarrow \quad & \frac{[ABC]}{s} = 2 \cdot \frac{abc}{4[ABC]} \cdot \frac{1 - \cos A}{2}.
\end{aligned}$$

Here, we have applied the area formulae and the half angle formula. Then

$$\begin{aligned}
& \angle AIO = 90^\circ \\
\Leftrightarrow \quad & 4[ABC]^2 = sabc(1 - \cos A) \\
\Leftrightarrow \quad & 4s(s-a)(s-b)(s-c) = sabc\left(1 - \frac{b^2 + c^2 - a^2}{2bc}\right) \\
\Leftrightarrow \quad & 8(s-a)(s-b)(s-c) = a(2bc - b^2 - c^2 + a^2) \\
\Leftrightarrow \quad & 8(s-a)(s-b)(s-c) = a\big(a^2 - (b-c)^2\big) \\
\Leftrightarrow \quad & 8(s-a)(s-b)(s-c) = 4a(s-b)(s-c) \\
\Leftrightarrow \quad & 2(s-a) = a \\
\Leftrightarrow \quad & b + c = 2a
\end{aligned}$$

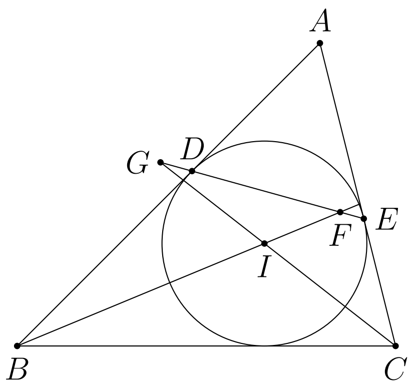
as desired. □

Remarks. The half angle formulae is commonly used in problems involving the incentre and the angle bisectors.

Problem 2.7. (CSMO 2006 Problem 5) In $\triangle ABC$, $\angle A = 60°$. The incircle $\odot I$ of $\triangle ABC$ touches the sides AB, AC at the points D, E respectively. The line DE intersects the lines BI, CI at the points F, G respectively. Prove that $FG = \frac{1}{2}BC$.

Analysis. There are many pairs of similar triangles and groups of concyclic points in the figure. This is the way of solving this problem in a geometric way. In case one cannot observe these, one might use trigonometric functions to calculate the desired lengths. For the segment FG, its length seems to be hard to find directly, as points F and G are not so related. One possible way is to find the lengths of other segments on EG.

Solution.



We first find the length of EG. In particular, we would use $\triangle CEG$ as all interior angles of this triangle can be found easily, and the length CE is also familiar to us. We know that $EC = s - c$, $\angle GCE = \frac{C}{2}$ and

$$\angle EGC = \angle DEA - \angle GCE = \left(90° - \frac{A}{2}\right) - \frac{C}{2} = \frac{B}{2}.$$

Using the sine law, we get

$$EG = EC \cdot \frac{\sin \angle GCE}{\sin \angle EGC} = \frac{(s-c)\sin\frac{C}{2}}{\sin\frac{B}{2}}. \tag{1}$$

Due to symmetry, we also have

$$DF = \frac{(s-b)\sin\frac{B}{2}}{\sin\frac{C}{2}}. \tag{2}$$

Next, as $\angle A = 60°$, $\triangle ADE$ is equilateral. This gives $DE = AD = AE = s - a$.

Together with (1) and (2), we have

$$FG = EG + DF - DE = \frac{(s-c)\sin\dfrac{C}{2}}{\sin\dfrac{B}{2}} + \frac{(s-b)\sin\dfrac{B}{2}}{\sin\dfrac{C}{2}} - (s-a).$$

It suffices to prove

$$\frac{(s-c)\sin\dfrac{C}{2}}{\sin\dfrac{B}{2}} + \frac{(s-b)\sin\dfrac{B}{2}}{\sin\dfrac{C}{2}} = \frac{b+c}{2}.$$

We first need to express $\sin\dfrac{B}{2}$ in terms of a, b, c. To do so, we use the half angle formula. Since $\sin\dfrac{B}{2} > 0$, we have

$$\begin{aligned}\sin\frac{B}{2} &= \sqrt{\frac{1-\cos B}{2}} = \sqrt{\frac{1-\dfrac{c^2+a^2-b^2}{2ca}}{2}} = \sqrt{\frac{b^2-(a-c)^2}{4ca}}\\ &= \sqrt{\frac{(b-a+c)(b+a-c)}{4ca}} = \sqrt{\frac{(s-a)(s-c)}{ac}}.\end{aligned}$$

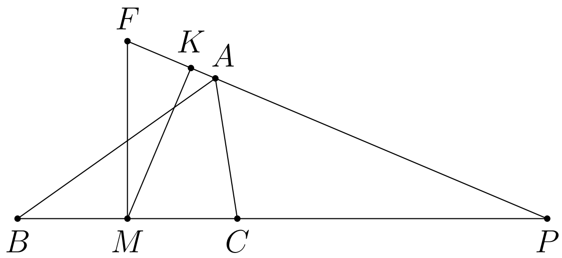
Similarly, $\sin\dfrac{C}{2} = \sqrt{\dfrac{(s-a)(s-b)}{ab}}$. This yields

$$\begin{aligned}\frac{(s-c)\sin\dfrac{C}{2}}{\sin\dfrac{B}{2}} + \frac{(s-b)\sin\dfrac{B}{2}}{\sin\dfrac{C}{2}} &= \frac{(s-c)\sin^2\dfrac{C}{2} + (s-b)\sin^2\dfrac{B}{2}}{\sin\dfrac{B}{2}\sin\dfrac{C}{2}}\\ &= \frac{\dfrac{(s-c)(s-a)(s-b)}{ab} + \dfrac{(s-b)(s-a)(s-c)}{ac}}{\sqrt{\dfrac{(s-a)(s-c)}{ac}}\cdot\sqrt{\dfrac{(s-a)(s-b)}{ab}}}\\ &= \frac{(s-b)(s-c)(b+c)}{\sqrt{bc(s-b)(s-c)}}\\ &= \sqrt{\frac{(s-b)(s-c)}{bc}}\cdot(b+c).\end{aligned}$$

To show that this is equal to $\dfrac{b+c}{2}$, it suffices to justify $\sqrt{\dfrac{(s-b)(s-c)}{bc}} = \dfrac{1}{2}$. This is true since $\sqrt{\dfrac{(s-b)(s-c)}{bc}} = \sin\dfrac{A}{2} = \sin 30^\circ = \dfrac{1}{2}$. $\square$

Remarks. The expression for $\sin\dfrac{A}{2}$ in terms of a, b, c is commonly used, in particular for problems involving the incentre. Note that the lengths $s-b$ and $s-c$, etc., are involved since $\dfrac{A}{2}$ is related to angle bisectors and the incentre.

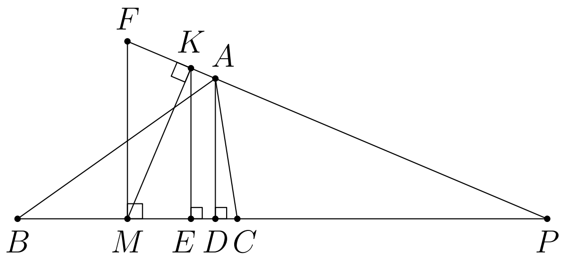
Problem 2.8. (CGMO 2010 Problem 6(2)) In acute $\triangle ABC$, $AB > AC$. Let M be the midpoint of side BC. The external angle bisector of $\angle BAC$ meets the line BC at P. Points K and F lie on the line PA such that $MF \perp BC$ and $MK \perp PA$. Prove that $BC^2 = 4PF \cdot AK$.



Analysis. The problem contains parts (1) and (2). The first part is a trivial geometric fact, so we do not bother to talk about it. For the result we are going to establish, our method is to convert the side lengths involved to a, b, c and prove the identity directly. So we only need to consider the lengths PF and AK.

As P, A, K, F all lie on the hypotenuse of a right-angled triangle, a natural way is to project these points on the line BC for comparison. Then we get a lot of right angles, and hence trigonometric functions can be used. In other words, the basic idea of using trigonometry is to construct more perpendicular lines.

Solution.



Let D and E be points on the line BC such that $AD \perp BC$ and $KE \perp BC$. Then we easily get $PF = \dfrac{PM}{\cos\angle APB}$ and $AK = \dfrac{DE}{\cos\angle APB}$. We need some more information about the location of P. By the external angle bisector theorem, we have $\dfrac{PB}{PC} = \dfrac{c}{b}$. This gives $\dfrac{CB}{PC} = \dfrac{c-b}{b}$, and hence $PC = \dfrac{ab}{c-b}$. Thus, we have

$$PM = PC + \frac{a}{2} = \frac{a(b+c)}{2(c-b)}. \tag{1}$$

Next, we have $PE = PK\cos\angle APB = PM\cos^2\angle APB$ using the right-angled triangles. Also, $CD = b\cos C = \dfrac{a^2+b^2-c^2}{2a}$. Therefore, we deduce

$$DE = PE - PC - CD = \frac{a(b+c)}{2(c-b)}\cos^2\angle APB - \frac{ab}{c-b} - \frac{a^2+b^2-c^2}{2a}. \tag{2}$$

From (1) and (2), we obtain

$$\begin{aligned}
&4PF\cdot AK\\
&= 4\cdot\frac{PM}{\cos\angle APB}\cdot\frac{DE}{\cos\angle APB}\\
&= \frac{4}{\cos^2\angle APB}\cdot\frac{a(b+c)}{2(c-b)}\left(\frac{a(b+c)}{2(c-b)}\cos^2\angle APB - \frac{ab}{c-b} - \frac{a^2+b^2-c^2}{2a}\right)\\
&= \frac{2a(b+c)}{c-b}\left(\frac{a(b+c)}{2(c-b)} - \frac{1}{\cos^2\angle APB}\cdot\frac{a^2b+a^2c+(b^2-c^2)(c-b)}{2a(c-b)}\right)\\
&= \frac{2a(b+c)}{c-b}\left(\frac{a(b+c)}{2(c-b)} - \frac{1}{\cos^2\angle APB}\cdot\frac{(b+c)\big(a^2-(b-c)^2\big)}{2a(c-b)}\right)\\
&= \frac{(b+c)^2}{(c-b)^2}\left(a^2 - \frac{a^2-(b-c)^2}{\cos^2\angle APB}\right).
\end{aligned} \tag{3}$$

Note that $\angle APB = C - \angle CAP = C + \dfrac{A}{2} - 90^\circ = \dfrac{C-B}{2}$. Therefore,

$$\begin{aligned}
&\cos^2\angle APB\\
&= \frac{1+\cos 2\angle APB}{2}\\
&= \frac{1+\cos(C-B)}{2}\\
&= \frac{1+\cos C\cos B+\sin C\sin B}{2}\\
&= \frac{1}{2}\left(1+\frac{a^2+b^2-c^2}{2ab}\cdot\frac{a^2+c^2-b^2}{2ac}+\frac{2[ABC]}{ab}\cdot\frac{2[ABC]}{ac}\right)\\
&= \frac{4a^2bc+(a^2+b^2-c^2)(a^2+c^2-b^2)+(-a^4-b^4-c^4+2a^2b^2+2b^2c^2+2c^2a^2)}{8a^2bc}\\
&= \frac{-2b^4+4b^2c^2-2c^4+2a^2b^2+4a^2bc+2a^2c^2}{8a^2bc}.
\end{aligned} \tag{4}$$

From (3) and (4), we get

$$\begin{aligned}
&4PF\cdot AK\\
&= \frac{(b+c)^2}{(c-b)^2}\left(a^2 - \frac{8a^2bc\big(a^2-(b-c)^2\big)}{-2b^4+4b^2c^2-2c^4+2a^2b^2+4a^2bc+2a^2c^2}\right)\\
&= \frac{a^2(b+c)^2\Big(\big(-2b^4+4b^2c^2-2c^4+2a^2b^2+4a^2bc+2a^2c^2\big)-8bc\big(a^2-b^2-c^2+2bc\big)\Big)}{(c-b)^2\big(-2b^4+4b^2c^2-2c^4+2a^2b^2+4a^2bc+2a^2c^2\big)}
\end{aligned}$$

$$= \frac{a^2(b+c)^2(-2b^4+8b^3c-12b^2c^2+8bc^3-2c^4+2a^2b^2-4a^2bc+2a^2c^2)}{(c-b)^2(-2b^4+4b^2c^2-2c^4+2a^2b^2+4a^2bc+2a^2c^2)}$$
$$= \frac{a^2(b+c)^2(c-b)^2(-2b^2+4bc-2c^2+2a^2)}{(c-b)^2(b+c)^2(-2b^2+4bc-2c^2+2a^2)} = a^2$$

as desired. Note that the factorization is trivial due to the result we want. □

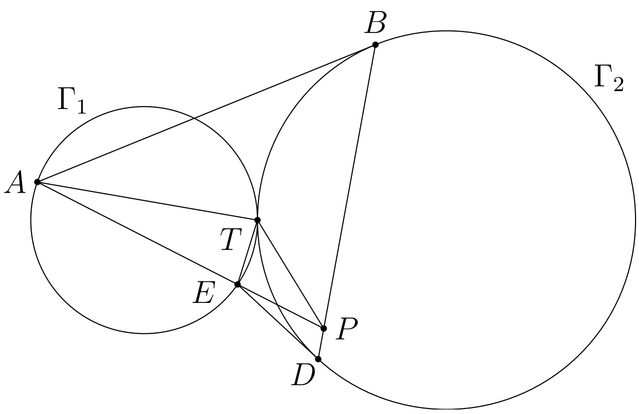
Remarks. This solution demonstrates one standard way of using trigonometric functions to solve triangle geometry problems. The goal is to express every length we want in terms of a, b, c. Sometimes we may need to use the formulae relating the trigonometric functions to rewrite them in terms of $\sin A, \cos B$, etc. These can then be converted to a, b, c via the sine law, the cosine formula and the area formulae. The way that we do in this solution is certainly not the best algebraic method. But it is for sure that one can do it along this way. In many cases, the factorizations involve symmetric terms like $b-c$ and $b+c$.

Even when we are doing the problem in an algebraic way, we can still use some geometric knowledge to simplify our calculations. For example, we may make use of some concyclic points in this problem to simplify the expansion steps.

Problem 2.9. (CGMO 2012 Problem 2) Circles Γ_1 and Γ_2 are tangent to each other externally at point T. Points A and E are on the circle Γ_1, lines AB and DE are tangent to the circle Γ_2 at points B and D respectively, lines AE and BD meet at point P. Prove that

(1) $\dfrac{AB}{AT} = \dfrac{ED}{ET}$;

(2) $\angle ATP + \angle ETP = 180°$.

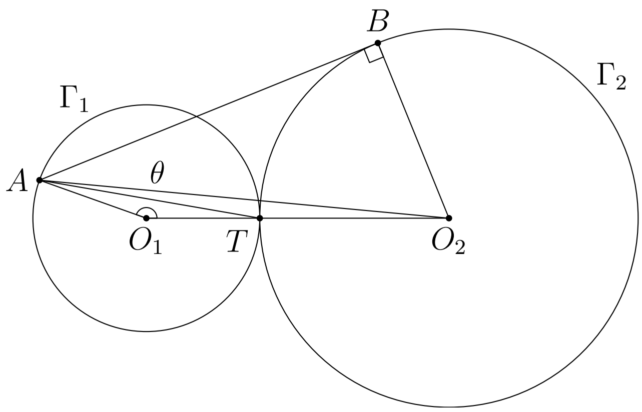


Analysis. For part (1), since the constructions for A, B and E, D are the same, we just need to show that the ratio $\dfrac{AB}{AT}$ is a constant, which means it only depends on the radii of the circles. Therefore, we would like to express everything in terms of these radii. A standard way is to connect the points on the circumferences to the centres.

For part (2), notice that the desired result is equivalent to $\sin\angle ATP = \sin\angle ETP$ (or $\cos\angle ATP + \cos\angle ETP = 0$) from their relative positions. We usually consider the sine function instead of the cosine function first, because the former usually does not involve radicals in applications. Also, the sine function is related to areas of triangles and the circumradii of triangles in a simple manner, hence is easier to handle.

Solution.

(1)



Suppose the centres and radii of Γ_1, Γ_2 are O_1, O_2 and r_1, r_2 respectively. Let $\angle AO_1O_2 = \theta$. We shall express $\dfrac{AB}{AT}$ in terms of r_1, r_2. Firstly,

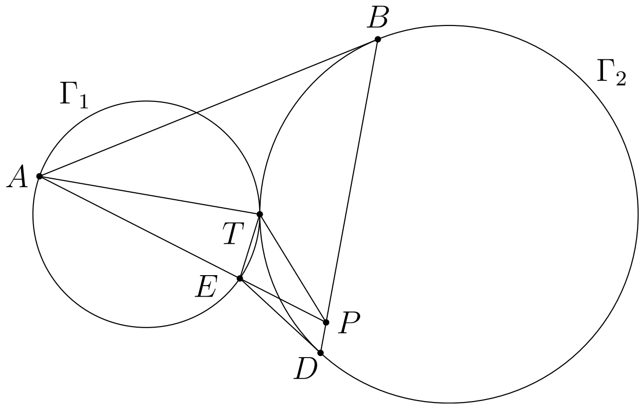
$$\begin{aligned} AB^2 = AO_2^2 - BO_2^2 &= (AO_1^2 + O_1O_2^2 - 2AO_1 \cdot O_1O_2 \cos\theta) - r_2^2 \\ &= r_1^2 + (r_1 + r_2)^2 - 2r_1(r_1 + r_2)\cos\theta - r_2^2 \\ &= 2r_1^2 + 2r_1r_2 - 2r_1(r_1 + r_2)\cos\theta \\ &= 2r_1(r_1 + r_2)(1 - \cos\theta). \end{aligned}$$

Secondly, we have

$$AT^2 = AO_1^2 + TO_1^2 - 2AO_1 \cdot TO_1 \cos\theta = r_1^2 + r_1^2 - 2r_1^2\cos\theta = 2r_1^2(1 - \cos\theta).$$

From these, we obtain $\dfrac{AB}{AT} = \sqrt{\dfrac{r_1 + r_2}{r_1}}$. Note that the above deduction does not require any information about the position of A on Γ_1. Therefore, we have $\dfrac{AB}{AT} = \dfrac{ED}{ET}$ by symmetry. □

(2) In the given configuration, we have $\angle ATP > \angle ETP$. Thus, these two angles cannot be equal. Then $\angle ATP + \angle ETP = 180°$ iff $\sin\angle ATP = \sin\angle ETP$. We use the area formula to express the sine functions, which can then be converted to other side lengths easily.



Indeed, from $\frac{1}{2}TA \cdot TP \sin\angle ATP = [TAP]$ and $\frac{1}{2}TE \cdot TP \sin\angle ETP = [TEP]$, we have $\sin\angle ATP = \sin\angle ETP$ iff $\frac{[TAP]}{TA} = \frac{[TEP]}{TE}$. As $\frac{[TAP]}{[TEP]} = \frac{AP}{EP}$ by proposition 2.1, this is equivalent to $\frac{TA}{TE} = \frac{AP}{EP}$. Then we can use the result of (1). Since $\frac{AB}{AT} = \frac{ED}{ET}$, i.e. $\frac{TA}{TE} = \frac{AB}{ED}$, it suffices to prove $\frac{AP}{EP} = \frac{AB}{ED}$.

Note that it is easier to relate AP and AB by the sine law, as well as EP and ED. Therefore, we rewrite

$$\begin{aligned} & \frac{AP}{EP} = \frac{AB}{ED} \\ \Leftrightarrow\quad & \frac{AP}{AB} = \frac{EP}{ED} \\ \Leftrightarrow\quad & \frac{\sin\angle ABP}{\sin\angle APB} = \frac{\sin\angle EDP}{\sin\angle EPD}. \end{aligned}$$

Since $\angle APB + \angle EPD = 180°$, we have $\sin\angle APB = \sin\angle EPD$. Also, we have $\angle ABP = \angle EDP$ since AB and ED are both tangent to Γ_2. This implies $\sin\angle ABP = \sin\angle EDP$. The proof is thus complete. □

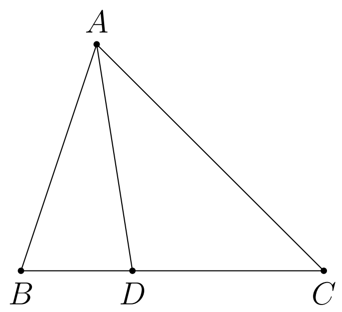
Remarks. In the solution of part (2), we mainly transform the desired expression to other forms through various relationship, including areas, the sine law and angle chasing. These techniques are quite standard for a solution involving trigonometry. The only goal is to convert everything to the simplest form.

For example, in the first step, we change the concerned angles to the side lengths TA, TE, PA, PE, which seem to be easier to deal with. Next, we notice that the point T is rather unfriendly. So we simply apply the result in (1) to eliminate the lengths involving T, and finally we get an expression which is obviously true.

In the solution of part (2) of the above problem, we find a relation between some angles and lengths in $\triangle TAP$ using the ratio of areas of triangles. The result can be summarized as follows.

Proposition 2.3. (Ratio lemma) In $\triangle ABC$, let D be a point on the sideline BC. Then

$$\frac{BD}{CD} = \frac{AB \sin \angle BAD}{AC \sin \angle CAD}.$$



Proof. By proposition 2.1, we have

$$\frac{BD}{CD} = \frac{[ABD]}{[ACD]} = \frac{\frac{1}{2}AB \cdot AD \sin \angle BAD}{\frac{1}{2}AC \cdot AD \sin \angle CAD} = \frac{AB \sin \angle BAD}{AC \sin \angle CAD}.$$

□

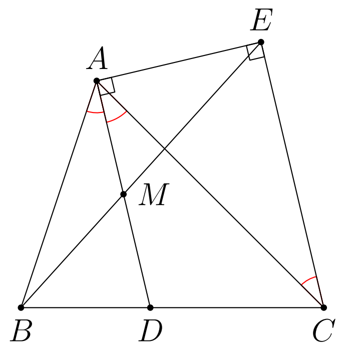
Note that this is a generalized result of the internal/external angle bisector theorem. The usefulness of this result is as follows. Suppose the point D is defined as the intersection point of BC and a line passing through A with a known direction. It is in general not easy to find the lengths BD and CD from the geometric perspective. The cosine formula does not work well because the expressions are ugly. However, the ratio lemma often provides a simple expression of $\frac{BD}{CD}$ to us. In particular, this is extremely useful in proving that a line bisects another line.

Problem 2.10. In $\triangle ABC$, the internal angle bisector of $\angle BAC$ intersects BC at D. Let E be a point such that $EC // AD$ and $AE \perp AD$. Prove that BE bisects AD.

Analysis. This is the situation as discussed. Let BE meet AD at M. It is not an easy task to find AM and DM separately. However, we are able to find $\frac{AM}{DM}$ using the ratio lemma efficiently. The rest is just to apply the sine law in different triangles.

Solution. Let BE meet AD at M. By the ratio lemma, we have

$$\frac{AM}{DM} = \frac{BA \sin \angle ABM}{BD \sin \angle DBM} = \frac{BA \sin \angle ABE}{BD \sin \angle CBE}.$$



The goal is to show that this ratio is equal to 1. The whole problem is now irrelevant to the point M. Note that $\angle AEC = 90^\circ$ since $EC // AD$ and $AE \perp AD$. To handle the terms $\sin \angle ABE$ and $\sin \angle CBE$, the only natural idea is to apply the sine law in $\triangle ABE$ and $\triangle CBE$, since these are the only triangles involving these angles (if the point M is removed). Indeed, we have

$$\frac{\sin \angle ABE}{AE} = \frac{\sin \angle BAE}{BE} = \frac{\sin \angle AEB}{AB},$$
$$\frac{\sin \angle CBE}{CE} = \frac{\sin \angle BCE}{BE} = \frac{\sin \angle BEC}{BC}.$$

Note that $\angle AEB$ and $\angle BEC$ are also hard to find. Thus, we should not consider the last ratio in each of the above expressions. We now find that

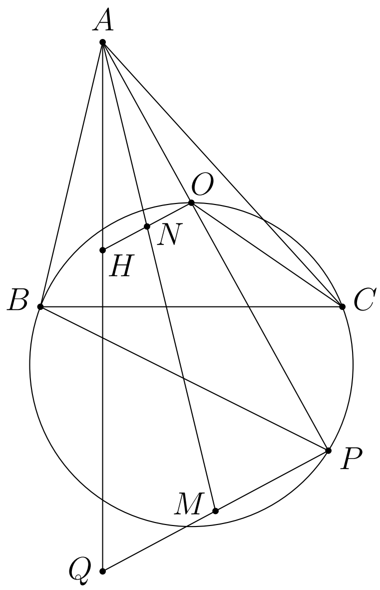
$$\begin{aligned}
\frac{BA \sin \angle ABE}{BD \sin \angle CBE} &= \frac{BA \cdot \frac{AE}{BE} \cdot \sin \angle BAE}{BD \cdot \frac{CE}{BE} \cdot \sin \angle BCE} \\
&= \frac{BA}{BD} \cdot \frac{AE}{CE} \cdot \frac{\sin \angle BAE}{\sin \angle BCE} \\
&= \frac{\sin \angle ADB}{\sin \angle BAD} \cdot \tan \angle ECA \cdot \frac{\sin \angle BAE}{\sin \angle BCE} \\
&= \frac{\sin\left(\frac{A}{2} + C\right)}{\sin \frac{A}{2}} \cdot \tan \frac{A}{2} \cdot \frac{\sin\left(90^\circ + \frac{A}{2}\right)}{\sin\left(\frac{A}{2} + C\right)} \\
&= \frac{\tan \frac{A}{2} \cos \frac{A}{2}}{\sin \frac{A}{2}} \\
&= 1.
\end{aligned}$$

This proves $\dfrac{AM}{DM} = 1$, and hence BE bisects AD. □

Problem 2.11. (Thailand TST 2014 Test 4 Problem 1) Let $\triangle ABC$ be an acute triangle with circumcentre O and nine-point centre N. Let P be the second intersection of AO and the circumcircle of $\triangle BOC$, and let Q be the reflection of A in BC. Show that the midpoint of segment PQ lies on line AN.

Analysis. We are going to use the ratio lemma to prove the result as before. Thus, we need some information about $\angle PAN$ and $\angle QAN$. Although these are not some angles which can be expressed in terms of A, B, C directly, we can find out the relation of their sine values by applying the ratio lemma in another triangle apart from $\triangle APQ$. One possibility is to consider $\triangle AHO$ where H is the orthocentre of $\triangle ABC$. The reason is that N is the midpoint of OH (which is the only geometric fact about N we need here), and we know very well how to compute AH and AO.

Solution.



Let H be the orthocentre of $\triangle ABC$. Note that H lies on AQ as both AH and AQ are perpendicular to BC. Let AN meet PQ at M. By the ratio lemma, we have

$$\frac{PM}{QM} = \frac{AP \sin \angle PAM}{AQ \sin \angle QAM}$$

and

$$\frac{AO \sin \angle OAN}{AH \sin \angle HAN} = \frac{ON}{HN} = 1.$$

Since $\angle PAM = \angle OAN$ and $\angle QAM = \angle HAN$, these yield

$$\frac{PM}{QM} = \frac{AP}{AQ} \cdot \frac{AH}{AO} = \frac{AP}{AQ} \cdot \frac{2R\cos A}{R}. \tag{1}$$

To find AP, we apply the sine law to $\triangle ABP$. This gives

$$AP = AB \cdot \frac{\sin \angle ABP}{\sin \angle APB} = \frac{c \sin(B + \angle POC)}{\sin \angle OCB} = \frac{c \sin(180^\circ - B)}{\sin(90^\circ - A)} = \frac{c \sin B}{\cos A}. \tag{2}$$

To find AQ, note that it is 2 times the height from A. Thus,

$$AQ = 2 \cdot \frac{2[ABC]}{BC} = \frac{4[ABC]}{a} = 2c \sin B. \tag{3}$$

Combining (1), (2) and (3), we deduce

$$\frac{PM}{QM} = \frac{1}{2\cos A} \cdot \frac{2R\cos A}{R} = 1.$$

This proves M is the midpoint of PQ as desired. □

Remarks. Those who have a better geometric insight should have realized that $\frac{AP}{AQ} = \frac{AO}{AH}$, and hence $AHNO \sim AQMP$. The above use of the ratio lemma can be avoided by using this similarity (or equivalently, the intercept theorem). But this is not something that one may observe at the beginning, especially for those who are not strong in geometry. Therefore, the above idea and solution seems to be more natural to us.

However, even in this computational approach, we still need to do some simple angle chasing (like the use of the concyclicity of C, O, B, P in the solution). One should always keep in mind (or mark in the figure) which angles are expressible in terms of A, B, C (such as $\angle ABP$) and which angles are not (such as $\angle OAN$).

We now deduce a useful corollary of the ratio lemma. Let $\theta = \angle BAC$ be a fixed angle. For any angle $x \in (0°, \theta)$, we can find a unique point D on BC such that $\angle BAD = x$. By the ratio lemma, we have

$$\frac{\sin\angle BAD}{\sin\angle CAD} = \frac{BD}{CD} \cdot \frac{AC}{AB}.$$

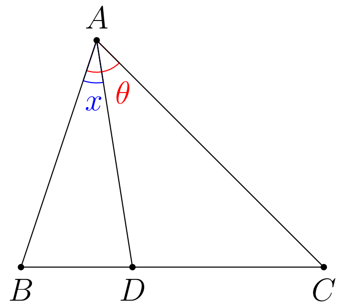
Since AB and AC are fixed, while $\frac{BD}{CD}$ is increasing when D moves from B to C, the function

$$f(x) = \frac{\sin x}{\sin(\theta - x)}$$

is strictly increasing. Therefore, it is injective. In other words, if we find that

$$\frac{\sin x}{\sin(\theta - x)} = \frac{\sin y}{\sin(\theta - y)}$$

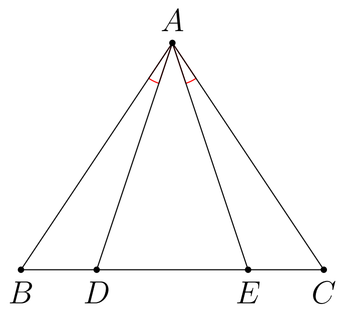
where $x, y \in (0°, \theta)$, then we must have $x = y$.



Problem 2.12. Let D and E be two points on the side BC of $\triangle ABC$. Suppose $BD = CE$ and $\angle BAD = \angle CAE$. Prove that $AB = AC$.

Analysis. The converse is trivial due to symmetry. However, we do not have enough information to prove $\triangle ABD \cong \triangle ACE$. Using trigonometry, we also end up with a relation of the sine values of some angles. We can use the injectivity of the function $f(x) = \dfrac{\sin x}{\sin(\theta - x)}$ to prove a pair of equal angles.

Solution.



Let $x = \angle ABC$, $y = \angle ACB$ and $\theta = \angle BAD = \angle CAE$. Applying the sine law in $\triangle ABD$ and $\triangle ACE$, we have

$$\frac{AD}{\sin x} = \frac{BD}{\sin\theta} = \frac{CE}{\sin\theta} = \frac{AE}{\sin y}.$$

This implies

$$\frac{\sin x}{\sin y} = \frac{AD}{AE} = \frac{\sin\angle AED}{\sin\angle ADE} = \frac{\sin(\theta + y)}{\sin(\theta + x)}.$$

Rearranging the terms, this gives

$$\frac{\sin x}{\sin(\theta + y)} = \frac{\sin y}{\sin(\theta + x)}.$$

Note that $x + (\theta + y) = y + (\theta + x) = 180^\circ - A + \theta$. Therefore, this means

$$\frac{\sin x}{\sin(180^\circ - A + \theta - x)} = \frac{\sin y}{\sin(180^\circ - A + \theta - y)},$$

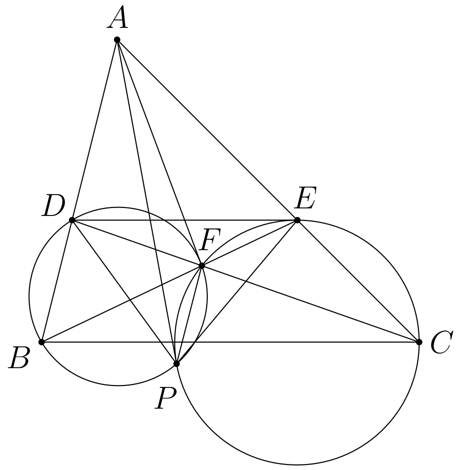
and hence $x = y$ as discussed. Thus, $AB = AC$. □

Problem 2.13. In $\triangle ABC$, D and E are points on the sides AB and AC respectively such that $DE // BC$. The segments BE and CD intersects at F. The circumcircles of $\triangle BDF$ and $\triangle CEF$ intersect again at P. Prove that $\angle BAP = \angle CAF$.

Analysis. It is not that easy to prove the equal angles because both angles cannot be easily expressed as the interior angles of $\triangle ABC$ (together with some angles depending on D and E). Note that the result basically means $\angle BAP = \angle CAF$

and $\angle CAP = \angle BAF$. As $\angle BAP + \angle CAP = \angle CAF + \angle BAF$, it suffices to prove $\dfrac{\sin\angle BAP}{\sin\angle CAP} = \dfrac{\sin\angle CAF}{\sin\angle BAF}$. This is easier to prove because we have some standard ways to handle ratio of sine values, and the expressions are symmetric. This is a standard way of proving equal angles if there is some sort of symmetry in the figure.

Solution.



Applying the sine law in $\triangle ADP$ and $\triangle AEP$, we have

$$\frac{\sin\angle BAP}{\sin\angle CAP} = \frac{\sin\angle ADP \cdot \dfrac{DP}{AP}}{\sin\angle AEP \cdot \dfrac{EP}{AP}} = \frac{\sin\angle BDP}{\sin\angle CEP}\cdot\frac{DP}{EP}$$
$$= \frac{\sin\angle BFP}{\sin\angle CFP}\cdot\frac{DP}{EP} = \frac{\sin\angle EFP}{EP}\cdot\frac{DP}{\sin\angle DFP}.$$

Using the extended sine law, this is equal to

$$\frac{\sin\angle CFE}{CE}\cdot\frac{BD}{\sin\angle BFD} = \frac{BD}{CE}.$$

Next, using the ratio lemma in $\triangle ACD$, we have

$$\frac{\sin\angle CAF}{\sin\angle BAF} = \frac{CF}{DF}\cdot\frac{AD}{AC}.$$

It remains to prove a simple relation

$$\frac{BD}{CE} = \frac{CF}{DF}\cdot\frac{AD}{AC}$$

under the assumption $DE//BC$. We do not bother to use trigonometry anymore, since the facts $\triangle ADE \sim \triangle ABC$ and $\triangle FED \sim \triangle FBC$ should be well-known to everyone. Using these similar triangles, we have

$$\frac{CF}{DF}\cdot\frac{AD}{AC} = \frac{BC}{DE}\cdot\frac{AD}{AC} = \frac{AC}{AE}\cdot\frac{AD}{AC} = \frac{AD}{AE} = \frac{BD}{CE}.$$

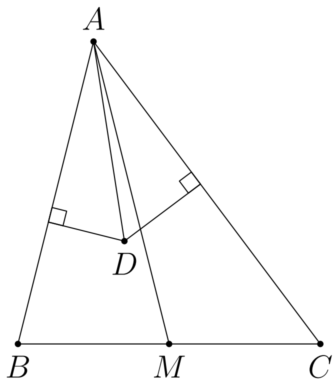
This proves $\dfrac{\sin\angle BAP}{\sin\angle CAP} = \dfrac{\sin\angle CAF}{\sin\angle BAF}$. As $\angle BAP + \angle CAP = \angle CAF + \angle BAF$, this implies $\angle BAP = \angle CAF$. $\square$

Lastly, we give a brief discussion on symmedians. Recall that the A-symmedian of $\triangle ABC$ is the reflection image of the A-median in the angle bisector of $\angle A$.

Instead of some standard geometric properties of the symmedians, we mainly focus on some properties related to the ratio of lengths of some segments.

Proposition 2.4. Let D be a point in $\angle BAC$. Let d_1 and d_2 be the distances from D to the lines AB and AC respectively. Then AD is the A-symmedian iff $\dfrac{d_1}{d_2} = \dfrac{AB}{AC}$ iff $\dfrac{\sin\angle BAD}{\sin\angle CAD} = \dfrac{AB}{AC}$.

In particular, if D lies on the side BC, then AD is the A-symmedian iff $\dfrac{BD}{CD} = \left(\dfrac{AB}{AC}\right)^2$.



Proof. Let M be the midpoint of BC. By the ratio lemma, we have

$$\frac{BM}{CM} = \frac{AB\sin\angle BAM}{AC\sin\angle CAM}.$$

As $BM = CM$, this yields

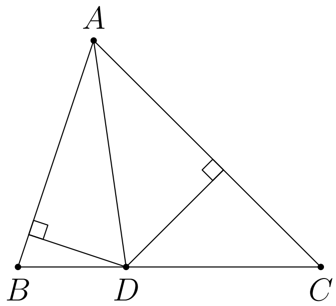
$$\frac{\sin\angle CAM}{\sin\angle BAM} = \frac{AB}{AC}.$$

On the other hand, using the right-angled triangles, we have

$$\frac{\sin\angle BAD}{\sin\angle CAD} = \frac{\frac{d_1}{AD}}{\frac{d_2}{AD}} = \frac{d_1}{d_2}.$$

It follows that

$$\begin{aligned}
& AD \text{ is the } A\text{-symmedian} \\
\Leftrightarrow\quad & \angle BAD = \angle CAM \\
\Leftrightarrow\quad & \frac{\sin\angle BAD}{\sin\angle CAD} = \frac{\sin\angle CAM}{\sin\angle BAM} \quad \text{(by the fact discussed above)} \\
\Leftrightarrow\quad & \frac{\sin\angle BAD}{\sin\angle CAD} = \frac{AB}{AC} \\
\Leftrightarrow\quad & \frac{d_1}{d_2} = \frac{AB}{AC}.
\end{aligned}$$



For the special case when D lies on BC, we have $d_1 = BD\sin B$ and $d_2 = CD\sin C$. Therefore,

$$\frac{d_1}{d_2} = \frac{BD\sin B}{CD\sin C} = \frac{BD}{CD}\cdot\frac{AC}{AB}.$$

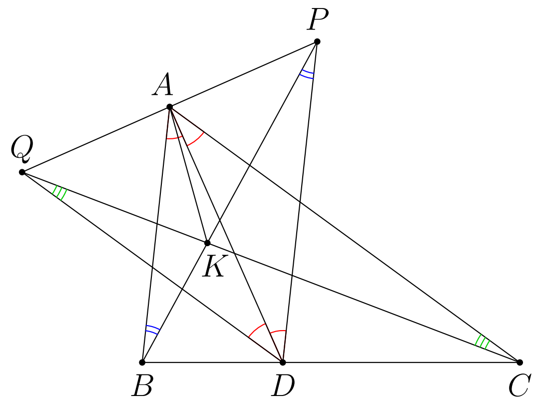
The equivalence follows immediately. □

We only provide one problem below to illustrate the use of this result.

Problem 2.14. (Sharygin Geometry Olympiad 2020 Correspondence Problem 11) Let ABC be a triangle with $\angle A = 60°$, AD be its bisector, and PDQ be a regular triangle with altitude DA. The lines PB and QC meet at point K. Prove that AK is a symmedian of ABC.

Analysis. Note that P and C should lie on the same side of the line AD, since otherwise PB and QC are parallel. This requires some proof, but we shall omit this in the following solution as this competition usually ignores the configuration issue. The only key idea in the proof is to compute $\dfrac{\sin\angle BAK}{\sin\angle CAK}$ in view of proposition 2.4. The rest is just to use the sine law in various triangles to rewrite everything in terms of the side lengths and interior angles of $\triangle ABC$.

Solution.



We are going to compute $\dfrac{\sin\angle BAK}{\sin\angle CAK}$. Applying the sine law in $\triangle ABK$ and $\triangle ACK$, we have

$$\frac{\sin\angle BAK}{\sin\angle CAK} = \frac{\sin\angle ABK \cdot \dfrac{BK}{AK}}{\sin\angle ACK \cdot \dfrac{CK}{AK}} = \frac{\sin\angle ABK}{\sin\angle ACK} \cdot \frac{BK}{CK}.$$

We choose not to use the ratios $\dfrac{AB}{\sin\angle AKB}$ and $\dfrac{AC}{\sin\angle AKC}$ because there is no obvious way to compute $\sin\angle AKB$ and $\sin\angle AKC$, even though the lengths AB and AC are simple. On the other hand, since $\angle BAD = \angle CAD = \angle ADP = \angle ADQ = 30°$, we have $AB // PD$ and $AC // QD$. Therefore, $\angle ABK = \angle BPD$ and $\angle ACK = \angle CQD$. Thus,

$$\begin{aligned}\frac{\sin\angle BAK}{\sin\angle CAK} &= \frac{\sin\angle ABK}{\sin\angle ACK} \cdot \frac{BK}{CK} = \frac{\sin\angle BPD}{\sin\angle CQD} \cdot \frac{\sin\angle BCK}{\sin\angle CBK} \\ &= \frac{\sin\angle BPD}{\sin\angle DBP} \cdot \frac{\sin\angle DCQ}{\sin\angle CQD} = \frac{BD}{PD} \cdot \frac{QD}{CD} = \frac{BD}{CD} = \frac{AB}{AC}.\end{aligned}$$

By proposition 2.4, we know that AK is a symmedian of $\triangle ABC$. □

Let us summarize the useful tools we have used in this section.

- trigonometric identities (problems 2.1, 2.4, 2.5, 2.6, 2.7, 2.8, 2.11)
- sine law (problems 2.1, 2.2, 2.3, 2.4, 2.5, 2.7, 2.9, 2.10, 2.11, 2.12, 2.13, 2.14)
- cosine formula (problems 2.4, 2.5, 2.6, 2.7, 2.8, 2.9)
- area formulae (problems 2.5, 2.6, 2.8, 2.11)
- ratio of areas/ratio lemma (problems 2.9, 2.10, 2.11, 2.13)
- Pythagoras' theorem (problems 2.5, 2.6)

- angle bisector theorem (problem 2.8)
- constructing right angles (problems 2.2, 2.5, 2.8)
- expressing everything in a, b, c or A, B, C (problems 2.5, 2.6, 2.7, 2.8, 2.10, 2.11)
- using injectivity of $\dfrac{\sin x}{\sin(\theta - x)}$ (problems 2.12, 2.13)
- ratio of lengths related to symmedians (problem 2.14)

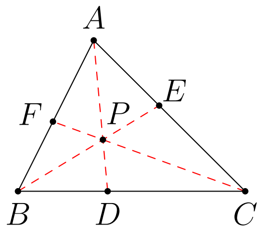
3 Concurrency and Collinearity

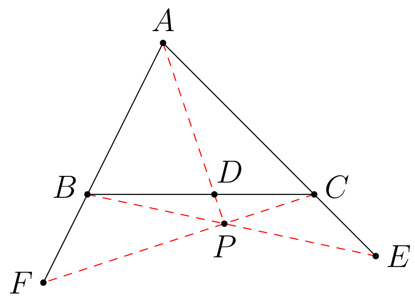
In this section, we mainly focus on some problems about concurrent lines and collinear points. These two kinds of problems are interchangeable by redefining the points. However, sometimes it is easier to tackle the problem in one way than another. Therefore, we can try both directions to see if they are easier to solve.

In this note, our main tools of proving concurrency and collinearity are Ceva's theorem and Menelaus' theorem.

Theorem 3.1. (Ceva's theorem(塞瓦定理)) In $\triangle ABC$, let D, E, F be points on the sidelines BC, CA, AB respectively. Then AD, BE, CF are concurrent iff

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1. \tag{1}$$





The lengths used in this theorem are directed lengths. Equivalently, one can only apply Ceva's theorem when 0 or 2 of the points D, E, F lie outside $\triangle ABC$, like the configurations shown above.

Proof. Suppose AD, BE, CF are concurrent at P. By proposition 2.1, we have

$$\frac{BD}{DC} = \frac{[ABP]}{[ACP]}.$$

Similarly, $\frac{CE}{EA} = \frac{[BCP]}{[ABP]}$ and $\frac{AF}{FB} = \frac{[ACP]}{[BCP]}$. Therefore, we have

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1.$$

Conversely, suppose (1) holds. Let BE meet CF at P, and let AP meet BC at D'. From above, as AD', BE, CF are concurrent, we have

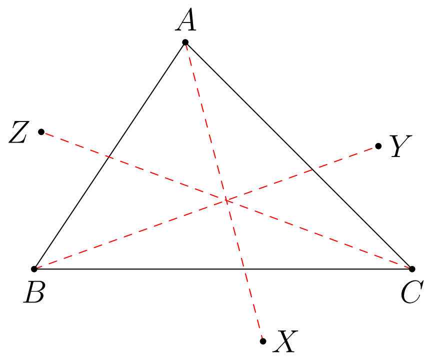
$$\frac{BD'}{D'C} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1. \tag{2}$$

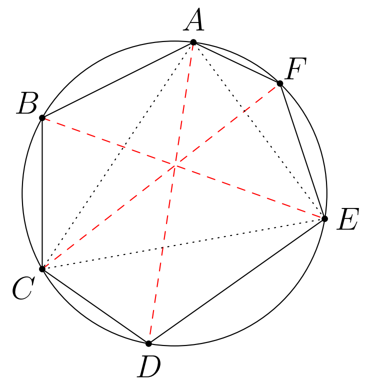
Comparing (1) and (2), we find that $\frac{BD}{DC} = \frac{BD'}{D'C}$. Since directed lengths are used, this forces $D = D'$, and hence AD, BE, CF are concurrent. $\square$

There is a trigonometric form of Ceva's theorem. Consider any distinct points A, B, C, X, Y, Z. Then AX, BY, CZ are concurrent iff

$$\frac{\sin \angle CAX}{\sin \angle XAB} \times \frac{\sin \angle ABY}{\sin \angle YBC} \times \frac{\sin \angle BCZ}{\sin \angle ZCA} = 1.$$

This follows easily by rewriting the fractions in Ceva's theorem using the ratio lemma. However, one has to be careful about the configuration, as using directed angles are not meaningful since the sine function is always positive on $(0°, 180°)$. In other words, the 'if' part should be carefully used. For example, if X, Y, Z lie inside $\angle BAC$, $\angle CBA$, $\angle ACB$ respectively, then we can use the result. This is the most common configuration that we have. This version of Ceva's theorem is the most useful one because we have more flexibility in choosing the points X, Y, Z. In particular, they need not lie on the sidelines of $\triangle ABC$.





In addition, we have another form of Ceva's theorem for points lying on a circle. Let A, B, C, D, E, F be any points on a circle in this order. Then AD, BE, CF are concurrent iff

$$\frac{AB}{BC} \times \frac{CD}{DE} \times \frac{EF}{FA} = 1.$$

To see this, we apply the trigonometric form of Ceva's theorem to $\triangle ACE$. The concurrency is equivalent to

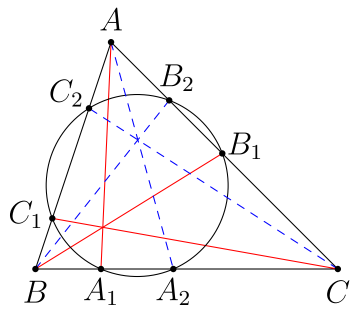
$$\frac{\sin \angle AEB}{\sin \angle CEB} \times \frac{\sin \angle CAD}{\sin \angle EAD} \times \frac{\sin \angle ECF}{\sin \angle ACF} = 1.$$

Using the extended sine law, we get the desired result.

Problem 3.1. (Japan EGMO Preliminary 2015 Problem 2) There is a triangle ABC, two different points A_1 and A_2 on the side BC, two different points B_1 and B_2 on the side CA, and different points C_1 and C_2 on the side AB, so that they are not endpoints. These 6 points are concyclic and AA_1, BB_1 and CC_1 are concurrent. Show that AA_2, BB_2 and CC_2 are concurrent.

Analysis. We use this problem to demonstrate the standard use of Ceva's theorem, both using it to obtain a relation of the lengths of some segments, and to prove the concurrency.

Solution.



By Ceva's theorem, since AA_1, BB_1, CC_1 are concurrent, we have

$$\frac{BA_1}{A_1C} \times \frac{CB_1}{B_1A} \times \frac{AC_1}{C_1B} = 1.$$

To prove that AA_2, BB_2, CC_2 are concurrent, it suffices to prove

$$\frac{BA_2}{A_2C} \times \frac{CB_2}{B_2A} \times \frac{AC_2}{C_2B} = 1$$

by Ceva's theorem. Thus, we need some relations between BA_1 and BA_2, etc. This suggests using the power of a point theorem. Indeed, from the concyclic points, we have

$$\begin{aligned} BA_1 \times BA_2 &= BC_1 \times BC_2, \\ CB_1 \times CB_2 &= CA_1 \times CA_2, \\ AC_1 \times AC_2 &= AB_1 \times AB_2. \end{aligned}$$

It follows that

$$\begin{aligned} \frac{BA_2}{A_2C} \times \frac{CB_2}{B_2A} \times \frac{AC_2}{C_2B} &= \frac{BA_2}{C_2B} \times \frac{CB_2}{A_2C} \times \frac{AC_2}{B_2A} \\ &= \frac{C_1B}{BA_1} \times \frac{A_1C}{CB_1} \times \frac{B_1A}{AC_1} \\ &= 1. \end{aligned}$$

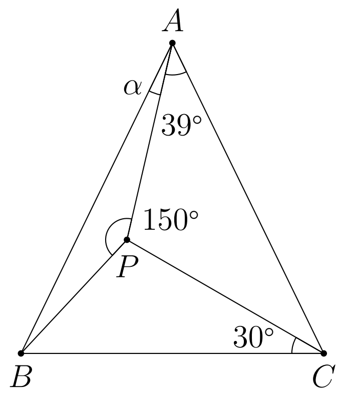
This proves AA_2, BB_2, CC_2 are concurrent. $\square$

Remarks. Let P be the concurrent point of AA_1, BB_1, CC_1, and let Q be the concurrent point of AA_2, BB_2, CC_2. Then Q is called the *cyclocevian conjugate*(圓塞瓦共軛) of P with respect to(w.r.t.) $\triangle ABC$.

Problem 3.2. (IMO Prelim 2014 Problem 9) $\triangle ABC$ is isosceles with $AB = AC$. P is a point inside $\triangle ABC$ such that $\angle BCP = 30°$, $\angle APB = 150°$ and $\angle CAP = 39°$. Find $\angle BAP$.

Analysis. Apart from the given angles, the isosceles condition and the angle sum of triangles, there is one more relation between the angles given by the trigonometric form of Ceva's theorem. This is because AP, BP, CP are concurrent at P. These relations are sufficient to deduce $\angle BAP$.

Solution.



Let $\alpha = \angle BAP$. Then $\angle ABP = 180° - 150° - \alpha = 30° - \alpha$. Note that $\triangle ABC$ is isosceles with

$$\angle ABC = \angle ACB = \frac{180° - \angle BAC}{2} = \frac{141° - \alpha}{2}.$$

This gives

$$\angle PBC = \angle ABC - \angle ABP = \frac{81° + \alpha}{2}$$

and

$$\angle ACP = \angle ACB - 30° = \frac{81° - \alpha}{2}.$$

Using the trigonometric form of Ceva's theorem, we obtain

$$\frac{\sin \angle PAC}{\sin \angle PAB} \times \frac{\sin \angle PBA}{\sin \angle PBC} \times \frac{\sin \angle PCB}{\sin \angle PCA} = 1.$$

We first note that

$$\begin{aligned}
& \frac{\sin 39°}{\sin \alpha} \times \frac{\sin(30° - \alpha)}{\sin \frac{81° + \alpha}{2}} \times \frac{\sin 30°}{\sin \frac{81° - \alpha}{2}} = 1 \\
\Rightarrow \quad & \frac{1}{2} \sin 39° \sin(30° - \alpha) = -\frac{1}{2} \sin \alpha (\cos 81° - \cos \alpha) \\
\Rightarrow \quad & \sin 39° \left(\frac{1}{2} \cos \alpha - \frac{\sqrt{3}}{2} \sin \alpha \right) = \sin \alpha \cos \alpha - \cos 81° \sin \alpha. \qquad (1)
\end{aligned}$$

We obtain an equation in α which we need to solve. The difficulty is that we do not know the exact values of $\sin 39°$ and $\cos 81°$. Therefore, we need to use a clever way to deal with these terms. These angles are not special, but one may note that they add up to a special angle $120°$. Therefore, we may express one of them in terms of the other. Indeed,

$$\cos 81° = \cos(120° - 39°) = -\frac{1}{2}\cos 39° + \frac{\sqrt{3}}{2}\sin 39°. \tag{2}$$

Putting (2) in (1) and simplifying, we get

$$\sin 39° \cos\alpha = 2\sin\alpha\cos\alpha + \cos 39° \sin\alpha,$$

i.e.

$$\sin 39° \cos\alpha - \cos 39° \sin\alpha = 2\sin\alpha\cos\alpha.$$

This implies

$$\sin(39° - \alpha) = \sin 2\alpha$$

Clearly, the configuration suggests $\alpha < 30°$. Therefore, both $39° - \alpha$ and 2α are acute. This forces $39° - \alpha = 2\alpha$, and hence $\alpha = 13°$. □

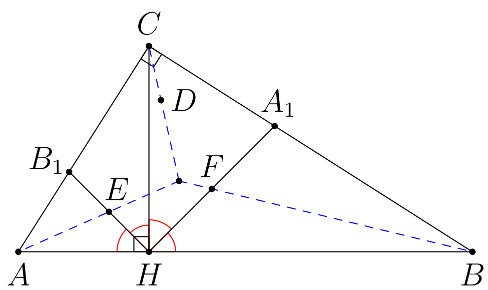
Remarks. There are many similar problems of this kind. The trigonometric form of Ceva's theorem together with the formulae introduced in section 1 can be used to solve most of those problems. However, one should be familiar with those trigonometric identities and use them in a clever way. Sometimes it is not entirely obvious how the equation can be solved.

The trigonometric form of Ceva's theorem can be used to solve Olympiad problems.

Problem 3.3. (Sharygin Geometry Olympiad 2021 Final Grade 10-11 Problem 1) Let CH be the altitude of a right-angled triangle ABC ($\angle C = 90°$), HA_1, HB_1 be the bisectors of angles CHB, AHC respectively, and E, F be the midpoints of HB_1 and HA_1 respectively. Prove that the lines AE and BF meet on the bisector of angle ACB.

Analysis. The trigonometric form of Ceva's theorem is suitable to prove the assertion since the lines AE, BF and the bisector of $\angle ACB$ are irrelevant, but we have the information about their directions.

Solution.



Let D be any point on the angle bisector of $\angle ACB$. By Ceva's theorem, it suffices to prove

$$\frac{\sin\angle ACD}{\sin\angle BCD}\times\frac{\sin\angle CBF}{\sin\angle ABF}\times\frac{\sin\angle BAE}{\sin\angle CAE}=1.$$

Clearly, the first fraction is 1 as $\angle ACD=\angle BCD$. For the second fraction, we use the ratio lemma to deduce

$$\frac{BA_1\sin\angle A_1BF}{BH\sin\angle HBF}=\frac{A_1F}{HF}=1.$$

This implies

$$\frac{\sin\angle CBF}{\sin\angle ABF}=\frac{BH}{BA_1}=\frac{\sin\angle BA_1H}{\sin\angle BHA_1}. \tag{1}$$

By symmetry,

$$\frac{\sin\angle BAE}{\sin\angle CAE}=\frac{\sin\angle AHB_1}{\sin\angle AB_1H}. \tag{2}$$

Note that $\angle BHA_1=\angle CHA_1=\angle CHB_1=\angle AHB_1=45^\circ$. Also, we have

$$\angle BA_1H=180^\circ-45^\circ-\angle A_1BH=135^\circ-B \tag{3}$$

and

$$\angle AB_1H=180^\circ-45^\circ-\angle B_1AH=135^\circ-A=135^\circ-(90^\circ-B)=45^\circ+B. \tag{4}$$

Combining (1), (2), (3) and (4), it follows that

$$\begin{aligned}\frac{\sin\angle ACD}{\sin\angle BCD}\times\frac{\sin\angle CBF}{\sin\angle ABF}\times\frac{\sin\angle BAE}{\sin\angle CAE}&=1\times\frac{\sin\angle BA_1H}{\sin\angle BHA_1}\times\frac{\sin\angle AHB_1}{\sin\angle AB_1H}\\&=\frac{\sin(135^\circ-B)}{\sin 45^\circ}\times\frac{\sin 45^\circ}{\sin(45^\circ+B)}\\&=1.\end{aligned}$$

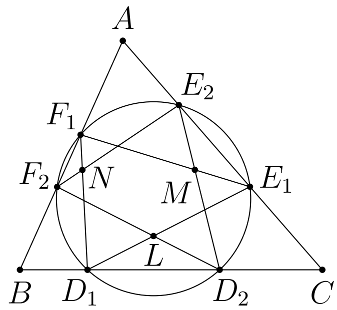
(Note that $135^\circ-B=180^\circ-(45^\circ+B)$.) This proves the assertion. □

Remarks. The last part can be simplified if we notice that $HAB_1C\sim HCA_1B$.

Problem 3.4. (CMO 2005 Problem 2) A circle intersects the sides BC, CA, AB of $\triangle ABC$ at D_1, D_2; E_1, E_2; F_1, F_2 in this order. The segments D_1E_1 and D_2F_2 intersect at a point L. The segments E_1F_1 and E_2D_2 intersect at a point M. The segments F_1D_1 and F_2E_2 intersect at a point N. Prove that AL, BM, CN are concurrent.

Analysis. To prove the concurrency, again we shall use Ceva's theorem. Since there is no triangle containing the three lines, it would be simpler to use the trigonometric form. To relate the angles $\angle BAL$ and $\angle CAL$, etc., we may simply use the sine law in different triangles. Although there are many points in the figure, it is not hard to figure out the correct way to rewrite the ratios.

Solution.



By Ceva's theorem, it suffices to prove

$$\frac{\sin\angle LAC}{\sin\angle LAB}\times\frac{\sin\angle MBA}{\sin\angle MBC}\times\frac{\sin\angle NCB}{\sin\angle NCA}=1$$

since all of L, M, N lie inside $\triangle ABC$. To find $\sin\angle LAC$, we consider $\triangle ALE_1$. This gives

$$\sin\angle LAC=\sin\angle AE_1D_1\cdot\frac{LE_1}{AL}. \tag{1}$$

By symmetry, we may consider $\triangle ALF_2$ to get

$$\sin\angle LAB=\sin\angle AF_2D_2\cdot\frac{LF_2}{AL}. \tag{2}$$

Combining (1) and (2), we get

$$\frac{\sin\angle LAC}{\sin\angle LAB}=\frac{\sin\angle AE_1D_1}{\sin\angle AF_2D_2}\cdot\frac{LE_1}{LF_2}.$$

Similarly,

$$\begin{aligned}&\frac{\sin\angle LAC}{\sin\angle LAB}\times\frac{\sin\angle MBA}{\sin\angle MBC}\times\frac{\sin\angle NCB}{\sin\angle NCA}\\ =&\frac{\sin\angle AE_1D_1}{\sin\angle AF_2D_2}\cdot\frac{LE_1}{LF_2}\times\frac{\sin\angle BF_1E_1}{\sin\angle BD_2E_2}\cdot\frac{MF_1}{MD_2}\times\frac{\sin\angle CD_1F_1}{\sin\angle CE_2F_2}\cdot\frac{ND_1}{NE_2}\\ =&\frac{LE_1}{LF_2}\times\frac{MF_1}{MD_2}\times\frac{ND_1}{NE_2}\end{aligned} \tag{3}$$

since $\angle AE_1D_1=\angle BD_2E_2$, etc. by the concyclic points. One may further use the sine law to simplify each fraction. For example, one may consider $\triangle LE_1D_2$ and $\triangle LF_2D_1$ to handle $\frac{LE_1}{LF_2}$. However, it is not hard to see that these triangles are indeed similar due to the elementary fact for concyclic points. No matter which method we use, we would obtain

$$\frac{LE_1}{LF_2}=\frac{D_2E_1}{D_1F_2}.$$

Multiplying similar equations, (3) is equal to

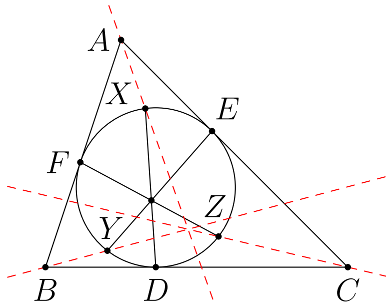
$$\frac{D_2E_1}{D_1F_2}\times\frac{E_2F_1}{E_1D_2}\times\frac{F_2D_1}{F_1E_2}=1.$$

This completes the proof. □

Problem 3.5. In $\triangle ABC$, the incircle touches the sides BC, CA, AB at D, E, F respectively. Let X, Y, Z be points on the minor arcs $\overset{\frown}{EF}, \overset{\frown}{FD}, \overset{\frown}{DE}$ of the incircle respectively such that DX, EY, FZ are concurrent. Prove that AX, BY, CZ are concurrent.

Analysis. This time we see an application of the circle form of Ceva's theorem. This allows us to use the concurrency directly.

Solution.



By Ceva's theorem, we need to prove

$$\frac{\sin\angle FAX}{\sin\angle EAX} \times \frac{\sin\angle ECZ}{\sin\angle DCZ} \times \frac{\sin\angle DBY}{\sin\angle FBY} = 1.$$

Applying the sine law in $\triangle AFX$ and $\triangle AEX$, we have

$$\frac{\sin\angle FAX}{\sin\angle EAX} = \frac{\sin\angle AFX \cdot \frac{FX}{AX}}{\sin\angle AEX \cdot \frac{EX}{AX}} = \frac{\sin\angle FEX}{\sin\angle EFX} \cdot \frac{FX}{EX} = \left(\frac{FX}{EX}\right)^2.$$

Similarly, we deduce

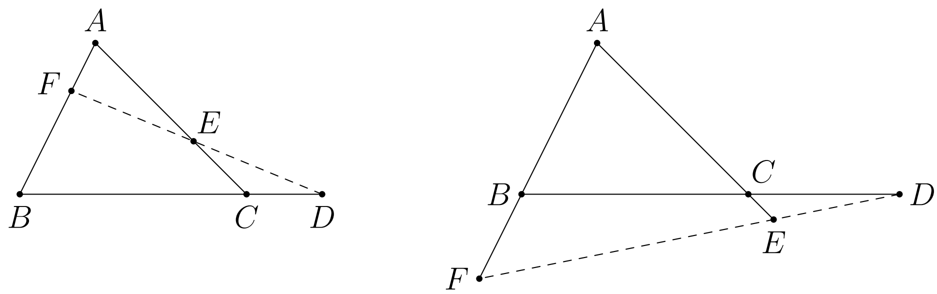
$$\frac{\sin\angle FAX}{\sin\angle EAX} \times \frac{\sin\angle ECZ}{\sin\angle DCZ} \times \frac{\sin\angle DBY}{\sin\angle FBY} = \left(\frac{FX}{EX} \times \frac{EZ}{DZ} \times \frac{DY}{FY}\right)^2 = 1.$$

The last equality holds by Ceva's theorem since the chords DX, EY, FZ are concurrent. □

Next, we move on to problems about collinearity.

Theorem 3.2. (Menelaus' theorem(梅涅劳斯定理)) In $\triangle ABC$, let D, E, F be points on the sidelines BC, CA, AB respectively. Then D, E, F are collinear iff

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = -1. \quad (1)$$



Again, the lengths are directed lengths. Equivalently, one can only apply Menelaus' theorem when 1 or 3 of the points D, E, F lie outside $\triangle ABC$, like the configurations shown above. In that case, the constant -1 can be regarded as 1.

Proof. Suppose D, E, F are collinear. By proposition 2.1, we have

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = \frac{[DBE]}{[DCE]} \times \frac{[DCE]}{[DAE]} \times \frac{[DAE]}{[DBE]} = 1.$$

Using directed lengths, the product should be -1 in view of the configuration.

Conversely, suppose (1) holds. Let FE meet BC again at D'. From above, as D', E, F are collinear, we have

$$\frac{BD'}{D'C} \times \frac{CE}{EA} \times \frac{AF}{FB} = -1. \tag{2}$$

Comparing (1) and (2), we find that $\frac{BD}{DC} = \frac{BD'}{D'C}$. Since directed lengths are used, this forces $D = D'$, and hence D, E, F are collinear. $\square$

There is also a trigonometric form of Menelaus' theorem. Using the same setting, D, E, F are collinear iff

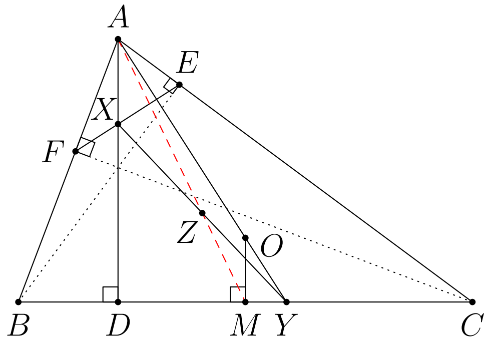
$$\frac{\sin \angle BAD}{\sin \angle DAC} \times \frac{\sin \angle CBE}{\sin \angle EBA} \times \frac{\sin \angle ACF}{\sin \angle FCB} = 1.$$

This again follows easily from the ratio lemma. However, this form is not that commonly used compared to the trigonometric form of Ceva's theorem. This is because we still require D, E, F to lie on the sidelines of $\triangle ABC$. This does not provide extra flexibility to us.

Problem 3.6. (Japan MO 2017 Problem 3) Let ABC be an acute-angled triangle with the circumcentre O. Let D, E and F be the feet of the altitudes from A, B and C, respectively, and let M be the midpoint of BC. AD and EF meet at X, AO and BC meet at Y, and let Z be the midpoint of XY. Prove that A, Z, M are collinear.

Analysis. We would like to use Menelaus' theorem to prove the collinearity. To do so, we have to find a triangle such that A, Z, M lie on the three sides of this triangle respectively. Based on the definition of Z, two of the vertices of this triangle must be X and Y. And then it is clear that the third vertex is $AX \cap MY = D$. Using Menelaus' theorem, it remains to prove a simple length relation.

Solution.



By Menelaus' theorem, A, Z, M are collinear iff

$$\frac{DA}{AX} \times \frac{XZ}{ZY} \times \frac{YM}{MD} = 1.$$

Note that usual lengths are used as only A lies outside $\triangle YDX$. It is given that $XZ = ZY$. It remains to express the other lengths in terms of the lengths and angles of $\triangle ABC$. Firstly, the height DA is equal to $b \sin C$. For AX, we can use the sine law in $\triangle AFX$. Since we do not know much about FX, we rewrite

$$AX = AF \cdot \frac{\sin \angle AFX}{\sin \angle AXF} = (b \cos A) \cdot \frac{\sin C}{\sin (180^\circ - \angle AFX - \angle FAX)} = \frac{b \cos A \sin C}{\sin (90^\circ - C + B)}$$

since it is well-known that $\angle AFX = \angle AFE = C$ and $\angle FAX = \angle BAD = 90^\circ - B$. This implies

$$\frac{DA}{AX} = \frac{\sin (90^\circ - C + B)}{\cos A}.$$

Next, for $\frac{YM}{MD}$, it is possible to compute YM and MD separately by using the positions of D, M, Y on the side BC. Here, we try a better approach. Indeed, the most troublesome point Y is defined as the intersection of AO and BC. Noting that $AD \perp BC$ and $OM \perp BC$, we may use the intercept theorem to deduce

$$\frac{YM}{MD} = \frac{YO}{OA}.$$

This kind of length chasing using the intercept theorem is also commonly used, especially when one is using Ceva's theorem and Menelaus' theorem (because all theorems involve the ratio of lengths on some straight lines). The expression on the right looks simpler since we already know $OA = R$. Indeed, we can do even better and get

$$\frac{YO}{OA} = \frac{YO}{OC} = \frac{\sin\angle OCY}{\sin\angle OYC} = \frac{\sin(90^\circ - A)}{\sin(\angle BAY + \angle ABY)} = \frac{\cos A}{\sin(90^\circ - C + B)}.$$

Combining all these, we find that

$$\frac{DA}{AX} \times \frac{XZ}{ZY} \times \frac{YM}{MD} = \frac{\sin(90^\circ - C + B)}{\cos A} \cdot 1 \cdot \frac{\cos A}{\sin(90^\circ - C + B)} = 1.$$

This proves A, Z, M are collinear. □

Remarks. For problems in triangle geometry, one has to be familiar with all simple angles and lengths related to the basic triangle centres.

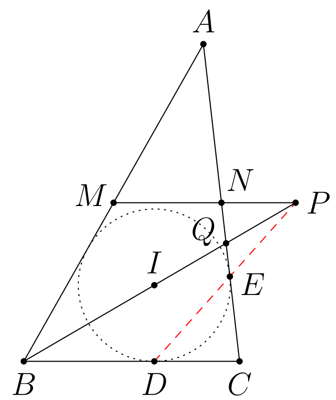
Problem 3.7. (Iran lemma(伊朗引理)) Let I be the incentre of $\triangle ABC$. The incircle of $\triangle ABC$ touches BC and AC at D and E respectively. Let M, N be the midpoints of the sides AB, AC respectively, and let P be the intersection point of the lines MN and BI. Prove that D, E, P are collinear.

Analysis. Let BI meet AC at Q. What we need to prove is

$$\frac{QP}{PB} \times \frac{BD}{DC} \times \frac{CE}{EQ} = 1.$$

All the lengths BD, DC, CE, EQ can be converted to a, b, c easily (of course under the assumption that we are familiar with the basic properties of the incentre and the angle bisectors). For $\frac{QP}{PB}$, by noting the definition of the point P, a simple way is to apply Menelaus' theorem again in $\triangle ABQ$. Once we have the correct plan, everything becomes a simple algebra problem.

Solution.



Let BI meet AC at Q. By considering $\triangle BCQ$, it suffices to prove

$$\frac{QP}{PB} \times \frac{BD}{DC} \times \frac{CE}{EQ} = 1 \tag{1}$$

by Menelaus' theorem (usual lengths can be used from the configuration). Firstly, we have $BD = s - b$ and $DC = CE = s - c$. By the angle bisector theorem, we have $\frac{CQ}{QA} = \frac{a}{c}$. It follows that

$$CQ = \frac{a}{a+c} \cdot AC = \frac{ab}{a+c}.$$

Therefore,

$$EQ = CQ - CE = \frac{ab}{a+c} - (s-c) = \frac{ab}{a+c} - \frac{a+b-c}{2} = \frac{(c-a)(a+c-b)}{2(a+c)}.$$

Next, applying Menelaus' theorem to $\triangle ABQ$ and the line PNM, we get

$$\frac{QP}{PB} \times \frac{BM}{MA} \times \frac{AN}{NQ} = 1. \tag{2}$$

Note that $BM = MA$, $AN = \frac{b}{2}$ and

$$NQ = CN - CQ = \frac{b}{2} - \frac{ab}{a+c} = \frac{b(c-a)}{2(a+c)}.$$

Putting everything into (2), we have

$$\frac{QP}{PB} \times \frac{\frac{b}{2}}{\frac{b(c-a)}{2(a+c)}} = 1.$$

Thus,

$$\frac{QP}{PB} = \frac{c-a}{a+c}.$$

Now, using all the results we have found, we obtain

$$\begin{aligned}
\frac{QP}{PB} \times \frac{BD}{DC} \times \frac{CE}{EQ} &= \frac{c-a}{a+c} \times \frac{s-b}{s-c} \times \frac{s-c}{\frac{(c-a)(a+c-b)}{2(a+c)}} \\
&= \frac{c-a}{a+c} \times \frac{s-b}{s-c} \times \frac{(s-c)(a+c)}{(c-a)(s-b)} \\
&= 1.
\end{aligned}$$

This establishes (1), and hence we are done. $\square$

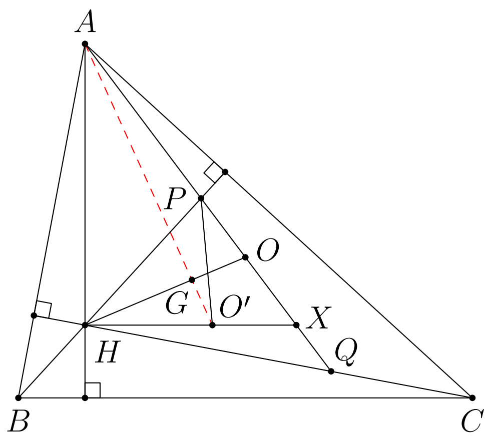
Problem 3.8. (IMO Shortlist 2017 G3) Let O be the circumcentre of an acute scalene triangle ABC. Line OA intersects the altitudes of ABC through B and C at P and Q, respectively. The altitudes meet at H. Prove that the circumcentre of triangle PQH lies on a median of triangle ABC.

Analysis. By symmetry, the circumcentre O' of $\triangle PQH$ should not lie on the B-median or the C-median (or it has to lie on both medians). Thus, we are going to prove that it lies on the A-median.

One difficulty in using Menelaus' theorem in this problem is that there is no straight line passing through O' using only the points given in the problem. Therefore, we have to define new points in order to use the theorem. Since O' is the circumcentre of $\triangle HPQ$, a natural idea is to use one of the lines HO', PO', QO'. The first choice looks better as we know more about the orthocentre H. Then it is natural to define $X = PQ \cap HO'$. We are going to apply Menelaus' theorem to a triangle with vertices H and X.

To show that O' lies on the A-median, the first approach is to show that it lies on AM where M is the midpoint of BC. However, this means the third vertex of the above triangle we are looking for should be $AH \cap XM$ or $AX \cap HM$. In either case, we do not have much information about the resultant triangle. The second approach is to show that it lies on AG where G is the centroid of $\triangle ABC$. This approach works much better, because G lies on OH in a known ratio. It remains to see how to use Menelaus' theorem to $\triangle OHX$.

Solution.



Let G be the centroid of $\triangle ABC$, let O' be the circumcentre of $\triangle HPQ$, and let X be the intersection point of HO' and AO. We need to show that A, G, O' are collinear. By Menelaus' theorem, this is equivalent to

$$\frac{HO'}{O'X} \times \frac{XA}{AO} \times \frac{OG}{GH} = 1.$$

It is well-known that $\frac{OG}{GH} = \frac{1}{2}$.

To compute the other two fractions, we need some information about $\triangle HPQ$. In fact, we are able to compute all its interior angles easily. This should be the first step one may attempt when working on this problem, so it is not surprising to obtain the following facts.

$$\begin{aligned}
&\angle HPQ = \angle BAP + \angle ABP = \angle BAO + \angle ABH = (90^\circ - C) + (90^\circ - A) = B,\\
&\angle HQP = C \quad \text{(by symmetry)},\\
&\angle PHQ = 180^\circ - \angle HPQ - \angle HQP = A.
\end{aligned}$$

Firstly, we can compute $\dfrac{HO'}{O'X}$ as in problem 3.6. This means

$$\frac{HO'}{O'X} = \frac{PO'}{O'X} = \frac{\sin\angle PXO'}{\sin\angle O'PX} = \frac{\sin(90^\circ - B + C)}{\sin(90^\circ - A)} = \frac{\cos(C-B)}{\cos A}.$$

Secondly, we may consider to use the sine law in $\triangle AHX$ to compute XA. This suggests finding $\angle AHX$ first. We find that

$$\angle AHX = \angle AHP + \angle PHO' = C + (90^\circ - C) = 90^\circ.$$

Since this is a right angle, we do not even need the sine law. Indeed, we plainly obtain

$$AX = \frac{AH}{\cos\angle HAX} = \frac{2R\cos A}{\cos\angle HAO} = \frac{2R\cos A}{\cos(B-C)}.$$

It follows that

$$\frac{HO'}{O'X} \times \frac{XA}{AO} \times \frac{OG}{GH} = \frac{\cos(C-B)}{\cos A} \times \frac{2\cos A}{\cos(B-C)} \times \frac{1}{2} = 1.$$

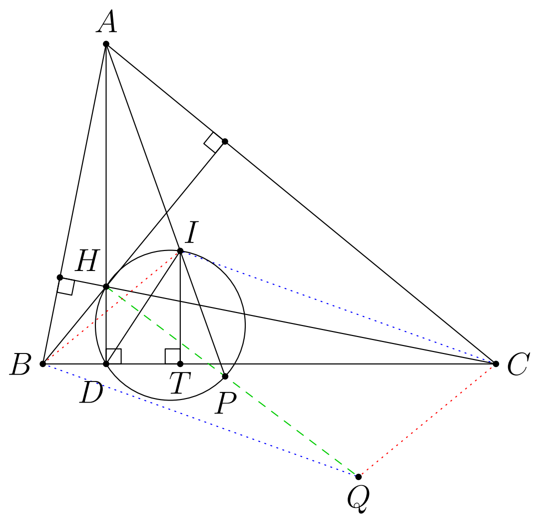
This proves A, G, O' are collinear by Menelaus' theorem, which means O' lies on the A-median. $\square$

Problem 3.9. In $\triangle ABC$, let H and I be its orthocentre and the incentre respectively. Let D be the foot of altitude from A. The circumcircle of $\triangle DHI$ meets the line AI again at P. Let Q be the point such that $CIBQ$ is a parallelogram. Prove that H, P, Q are collinear.

Analysis. This problem is also about proving collinearity. Instead of using Menelaus' theorem, we demonstrate how to use Ceva's theorem to prove the assertion. Indeed, it suffices to rephrase the assertion in terms of a problem about proving concurrency. For example, we can prove that BQ (the line through B parallel to IC), CQ (the line through C parallel to IB) and HP are concurrent.

To use Ceva's theorem, we can either use $\triangle BCH$ or $\triangle BCP$ (there is no other better option because B and C are the only points on BQ and CQ apart from the strange point Q). Obviously, we should use $\triangle BCH$. One can work out an equivalent statement to prove in terms of the angles. The remaining task is to transform everything to a, b, c and A, B, C in whatever way we can think of. The approach used in the solution below does not seem to be the best way to do this task, but it works anyway.

Solution.



WLOG assume $AB \leqslant AC$. It is the same as proving BQ, CQ and HP are concurrent. By the trigonometric form of Ceva's theorem, this is the same as

$$\frac{\sin \angle BHP}{\sin \angle CHP} \times \frac{\sin \angle HCQ}{\sin \angle BCQ} \times \frac{\sin \angle CBQ}{\sin \angle HBQ} = 1.$$

The last two fractions are easier to handle. Indeed, we have

$$\angle HCQ = \angle HCB + \angle BCQ = (90^\circ - B) + \frac{B}{2} = 90^\circ - \frac{B}{2}.$$

This yields

$$\frac{\sin \angle HCQ}{\sin \angle BCQ} = \frac{\sin\left(90^\circ - \frac{B}{2}\right)}{\sin \frac{B}{2}} = \frac{\cos \frac{B}{2}}{\sin \frac{B}{2}} = \frac{1}{\tan \frac{B}{2}}.$$

Similarly,

$$\frac{\sin \angle CBQ}{\sin \angle HBQ} = \tan \frac{C}{2}.$$

Next, we have

$$\begin{aligned} \angle BHP &= \angle BHD + \angle DHP = C + \angle DIP = C + \angle DAI + \angle ADI \\ &= C + \frac{B - C}{2} + \angle ADI = \frac{B + C}{2} + \angle ADI \end{aligned}$$

and

$$\angle CHP = \angle CHD - \angle DHP = B - \frac{B - C}{2} - \angle ADI = \frac{B + C}{2} - \angle ADI.$$

Therefore, we find that

$$\frac{\sin\angle BHP}{\sin\angle CHP} = \frac{\sin\frac{B+C}{2}\cos\angle ADI + \cos\frac{B+C}{2}\sin\angle ADI}{\sin\frac{B+C}{2}\cos\angle ADI - \cos\frac{B+C}{2}\sin\angle ADI} = \frac{\tan\frac{B+C}{2} + \tan\angle ADI}{\tan\frac{B+C}{2} - \tan\angle ADI}.$$

Note that

$$\tan\frac{B+C}{2} = \tan\left(90^\circ - \frac{A}{2}\right) = \frac{1}{\tan\frac{A}{2}} = \frac{s-a}{r}.$$

To compute $\tan\angle ADI$, let T be the foot of perpendicular from I to BC. This means T is the contact point of the incircle of $\triangle ABC$ with BC. Thus,

$$\begin{aligned}\tan\angle ADI = \tan\angle DIT = \frac{DT}{IT} &= \frac{BT - BD}{r} = \frac{(s-b) - c\cos B}{r}\\ &= \frac{a(a+c-b) - (a^2+c^2-b^2)}{2ar} = \frac{(b-c)(b+c-a)}{2ar} = \frac{(b-c)(s-a)}{ar}.\end{aligned}$$

It follows that

$$\frac{\sin\angle BHP}{\sin\angle CHP} = \frac{\dfrac{s-a}{r} + \dfrac{(b-c)(s-a)}{ar}}{\dfrac{s-a}{r} - \dfrac{(b-c)(s-a)}{ar}} = \frac{a+b-c}{a-b+c} = \frac{s-c}{s-b},$$

and hence

$$\frac{\sin\angle BHP}{\sin\angle CHP} \times \frac{\sin\angle HCQ}{\sin\angle BCQ} \times \frac{\sin\angle CBQ}{\sin\angle HBQ} = \frac{s-c}{s-b} \times \frac{1}{\tan\frac{B}{2}} \times \tan\frac{C}{2} = 1$$

since $\tan\dfrac{B}{2} = \dfrac{r}{s-b}$ and $\tan\dfrac{C}{2} = \dfrac{r}{s-c}$. This completes the proof. □

This section is mainly about proving concurrency and collinearity. Again, one should keep in mind that these two kinds of problems are equivalent to each other.

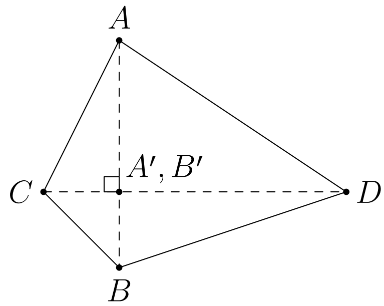
- Ceva's theorem
  - standard form (problem 3.1)
  - trigonometric form (problems 3.2, 3.3, 3.4, 3.5, 3.9)
  - circle form (problem 3.5)
- Menelaus' theorem (problems 3.6, 3.7, 3.8)

4 Not-So-Geometry Theorems

In this section, we introduce some more theorems in plane geometry which provide some relations in lengths. These results can be combined with the tools introduced in previous sections to solve more problems.

Theorem 4.1. Let A, B, C, D be distinct points on the plane. Then $AB \perp CD$ iff

$$AC^2 - AD^2 = BC^2 - BD^2. \tag{1}$$



Proof. Let A', B' be the projections of A, B on the line CD respectively. Then $AB \perp CD$ iff $A' = B'$. Note that

$$\begin{aligned}
& AC^2 - AD^2 = BC^2 - BD^2 \\
\Leftrightarrow\quad & (A'A^2 + A'C^2) - (A'A^2 + A'D^2) = (B'B^2 + B'C^2) - (B'B^2 + B'D^2) \\
\Leftrightarrow\quad & A'C^2 - A'D^2 = B'C^2 - B'D^2 \\
\Leftrightarrow\quad & (A'C - A'D)(A'C + A'D) = (B'C - B'D)(B'C + B'D) \\
\Leftrightarrow\quad & DC(DC + 2A'D) = DC(DC + 2B'D) \\
\Leftrightarrow\quad & DC + 2A'D = DC + 2B'D \\
\Leftrightarrow\quad & A'D = B'D \\
\Leftrightarrow\quad & A' = B'.
\end{aligned}$$

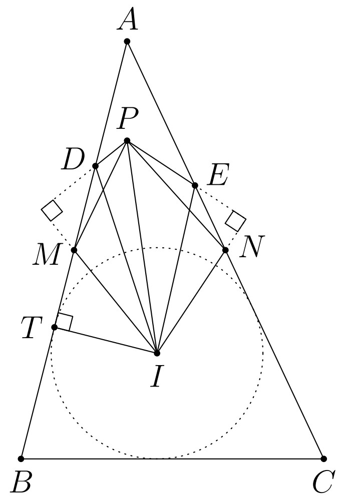
The result follows. (Note that we are using directed lengths in the proof.) □

This is a generalization of Pythagoras' theorem. It is more useful than the triangular form in proving a pair of perpendicular lines since the points A, B, C, D are rather arbitrary.

Problem 4.1. (China TST 2007 Quiz 1 Problem 2) Let I be the incentre of triangle ABC. Let M, N be the midpoints of AB, AC, respectively. Points D, E lie on AB, AC respectively such that $BD = CE = BC$. The line perpendicular to IM through D intersects the line perpendicular to IN through E at P. Prove that $AP \perp BC$.

Analysis. We demonstrate how to use theorem 4.1 to prove the perpendicularity. We just need to do step by step to express every length in terms of a, b, c. Noting that $BC//MN$, we shall prove $AP \perp MN$ instead, since the point P is more related to the points M and N.

Solution.



By the midpoint theorem, we have $MN//BC$. So it is the same as proving $AP \perp MN$. By theorem 4.1, we need to prove

$$PM^2 - PN^2 = AM^2 - AN^2. \tag{1}$$

Firstly, since $PD \perp MI$, we have

$$PM^2 - PI^2 = DM^2 - DI^2. \tag{2}$$

The length DM is easy to find. In fact, we have

$$DM = BD - BM = a - \frac{c}{2}.$$

For DI, we let T be the contact point of the incircle and the side AB. Then we have

$$DI^2 = DT^2 + r^2 = (BD - BT)^2 + r^2 = \left(a - (s - b)\right)^2 + r^2.$$

Putting these into (2), we find that

$$PM^2 = PI^2 + \left(a - \frac{c}{2}\right)^2 - (a + b - s)^2 - r^2.$$

By symmetry, we obtain

$$PN^2 = PI^2 + \left(a - \frac{b}{2}\right)^2 - (a + c - s)^2 - r^2.$$

Therefore,

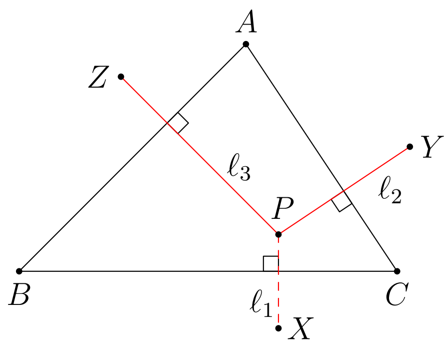
$$\begin{aligned}
& PM^2 - PN^2 = AM^2 - AN^2 \\
\Leftrightarrow \quad & \left(a - \frac{c}{2}\right)^2 - (a+b-s)^2 - \left(a - \frac{b}{2}\right)^2 + (a+c-s)^2 = \left(\frac{c}{2}\right)^2 - \left(\frac{b}{2}\right)^2 \\
\Leftrightarrow \quad & \left(\left(a - \frac{c}{2}\right)^2 - \left(\frac{c}{2}\right)^2\right) - \left(\left(a - \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2\right) = (a+b-s)^2 - (a+c-s)^2 \\
\Leftrightarrow \quad & a(a-c) - a(a-b) = a(b-c).
\end{aligned}$$

This clearly holds. So (1) is established, and we are done. □

Using the generalization of Pythagoras' theorem, we can prove another result about perpendicular lines.

Theorem 4.2. (Carnot's theorem(卡諾定理)) Let A, B, C, X, Y, Z be distinct points. Let ℓ_1 be the line through X perpendicular to BC, let ℓ_2 be the line through Y perpendicular to CA, and let ℓ_3 be the line through Z perpendicular to AB. Then ℓ_1, ℓ_2, ℓ_3 are concurrent iff

$$BX^2 - XC^2 + CY^2 - YA^2 + AZ^2 - ZB^2 = 0.$$



Proof. Let ℓ_2 and ℓ_3 meet at P. Then ℓ_1, ℓ_2, ℓ_3 are concurrent iff $XP \perp BC$. Since $YP \perp CA$ and $ZP \perp AB$, by theorem 4.1, we have

$$CY^2 - YA^2 = CP^2 - PA^2$$

and

$$AZ^2 - ZB^2 = AP^2 - PB^2.$$

Adding these, we get

$$CY^2 - YA^2 + AZ^2 - ZB^2 = CP^2 - PB^2.$$

Similarly, $XP \perp BC$ iff

$$BX^2 - XC^2 = BP^2 - PC^2.$$

Therefore, this is equivalent to

$$BX^2 - XC^2 + CY^2 - YA^2 + AZ^2 - ZB^2 = 0.$$

This completes the proof. $\square$

As a corollary, the following two statements are equivalent.

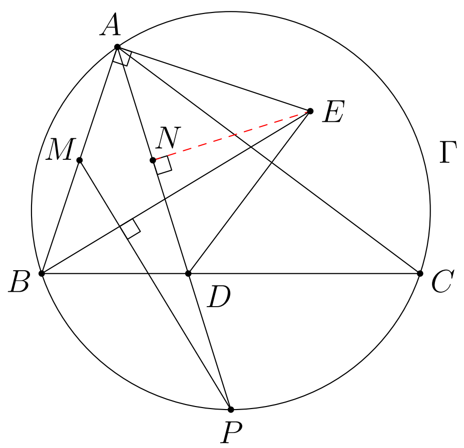
- The line through X perpendicular to BC, the line through Y perpendicular to CA and the line through Z perpendicular to AB are concurrent.
- The line through A perpendicular to YZ, the line through B perpendicular to ZX and the line through C perpendicular to XY are concurrent.

Carnot's theorem can be used to prove concurrent lines, where the lines are defined via some perpendicular conditions. It is not commonly used in practice because we need a lot of perpendicular relations in order to use the theorem. Sometimes we have to define extra points in an artificial way.

Problem 4.2. (IMO HK TST 2021 Test 2 Problem 3) Let $\triangle ABC$ be an acute triangle with circumcircle Γ, and let P be the midpoint of the minor arc BC of Γ. Let AP meet BC at D, and let M be the midpoint of AB. Also, let E be the point such that $AE \perp AB$ and $BE \perp MP$. Prove that $AE = DE$.

Analysis. There are two perpendicular conditions in the question. In particular, the condition $BE \perp MP$ is hard to use. This motivates the consideration of Carnot's theorem. The assertion $AE = DE$ is equivalent to E lying on the perpendicular bisector of AD. Therefore, we shall prove that AE, BE and the perpendicular bisector of AD are concurrent. This looks more similar to Carnot's theorem as the three lines involved are perpendicular to some lines in the figure. Indeed, they are perpendicular to AB, MP, AD respectively. These three lines enclose $\triangle AMP$, which suggests how Carnot's theorem should be applied.

Solution.



Let N be the midpoint of AD. We are going to apply Carnot's theorem to $\triangle ABN$ and $\triangle PAM$. Thus, we first prove

$$BP^2 - PN^2 + NA^2 - AA^2 + AM^2 - MB^2 = 0. \tag{1}$$

Note that $AA = 0$ and $AM^2 = MB^2$. Therefore, it remains to prove

$$BP^2 - PN^2 + NA^2 = 0.$$

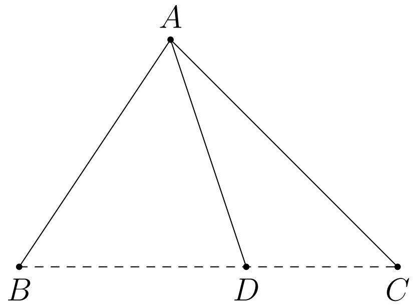
As N is the midpoint of AD, we have

$$PN^2 - NA^2 = (PN - NA)(PN + NA) = (PN - ND)(PN + NA) = PD \times PA.$$

This is equal to PB^2 since $\triangle PBD \sim \triangle PAB$. (If one fails to recall this simple fact, just use the sine law to prove this.) Therefore, (1) is established. By Carnot's theorem, the line through A perpendicular to AM, the line through B perpendicular to MP and the line through N perpendicular to PA are concurrent. The first two lines intersect at E by definition. Therefore, E lies on the perpendicular bisector of AD. This implies $AE = DE$. □

Theorem 4.3. (Subtended angle theorem(張角定理)) Let D be a point in the interior of $\angle BAC$. Then D lies on the segment BC iff

$$\frac{\sin\angle BAD}{AC} + \frac{\sin\angle CAD}{AB} = \frac{\sin\angle BAC}{AD}.$$



Proof. Indeed, D lies on the segment BC iff $[ABD] + [ACD] = [ABC]$. This is equivalent to

$$\frac{1}{2}AB\cdot AD\sin\angle BAD + \frac{1}{2}AC\cdot AD\sin\angle CAD = \frac{1}{2}AB\cdot AC\sin\angle BAC.$$

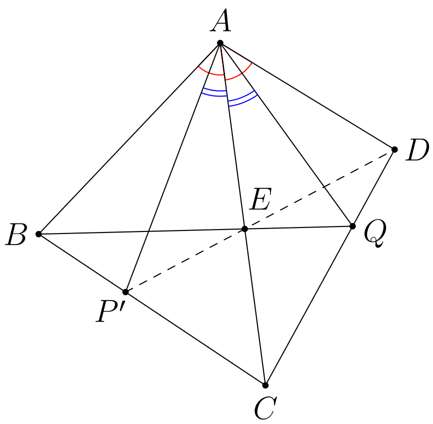
Dividing both sides by $\frac{1}{2}AB\cdot AC\cdot AD$, we get the desired equivalence. □

The subtended angle theorem gives another way to show collinearity. This is useful if we have more information on the angles $\angle BAD$ and $\angle CAD$.

Problem 4.3. In a convex quadrilateral $ABCD$, the diagonal AC bisects $\angle BAD$. Let E be a point on AC. The extension of DE meets the side BC at P. The extension of BE meets the side CD at Q. Prove that $\angle PAC = \angle QAC$.

Analysis. This is a special case of the isogonal line lemma, and is a classical example that can be solved by the subtended angle theorem. Instead of proving the equal angles, we use false position to prove the collinearity instead. In that case, we have three pairs of equal angles to work on.

Solution.



Let P' be the point on BC such that $\angle P'AC = \angle QAC = \alpha$. Then we need to prove P', E, D are collinear. Note that

$$\angle BAP' = \angle BAC - \angle P'AC = \angle DAC - \angle QAC = \angle DAQ.$$

Let $\angle BAP' = \angle DAQ = \beta$. Then $\angle BAC = \angle DAC = \alpha + \beta$. Using the subtended angle theorem on $\triangle ABC$ and the line $BP'C$; on $\triangle ACD$ and the line CQD; and on $\triangle ABQ$ and the line BEQ, we get

$$\frac{\sin\beta}{AC} + \frac{\sin\alpha}{AB} = \frac{\sin(\alpha+\beta)}{AP'}, \tag{1}$$

$$\frac{\sin\beta}{AC} + \frac{\sin\alpha}{AD} = \frac{\sin(\alpha+\beta)}{AQ}, \tag{2}$$

$$\frac{\sin\alpha}{AB} + \frac{\sin(\alpha+\beta)}{AQ} = \frac{\sin(2\alpha+\beta)}{AE} \tag{3}$$

respectively. Computing (2) + (3) − (1), we get

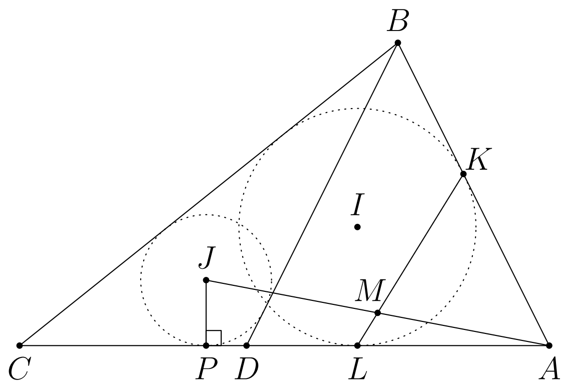
$$\frac{\sin\alpha}{AD} + \frac{\sin(\alpha+\beta)}{AP'} = \frac{\sin(2\alpha+\beta)}{AE}.$$

By the subtended angle theorem, this implies P', E, D are collinear as desired. □

Problem 4.4. (IMO Shortlist 2006 G4) A point D is chosen on the side AC of a triangle ABC with $\angle C < \angle A < 90°$ in such a way that $BD = BA$. The incircle of ABC is tangent to AB and AC at points K and L, respectively. Let J be the incentre of triangle BCD. Prove that the line KL intersects the line segment AJ at its midpoint.

Analysis. To prove the assertion, we simply need to calculate the lengths of AM and AJ, where M is the intersection of KL and AJ. From the configuration, the easiest way to find AM is to apply the subtended angle theorem, since the direction of the line AJ is fixed and we have no simple information about the segments MK and ML.

Solution.



Let KL meet AJ at point M, and let P be the projection from J on CA. Applying the subtended angle theorem to $\triangle AKL$ and the line KML, we get

$$\frac{\sin \angle KAM}{AL} + \frac{\sin \angle LAM}{AK} = \frac{\sin \angle KAL}{AM}.$$

This gives

$$\frac{\sin(A - \angle JAP)}{s-a} + \frac{\sin \angle JAP}{s-a} = \frac{\sin A}{AM},$$

and so

$$\begin{aligned} AM &= \frac{(s-a)\sin A}{\sin A \cos \angle JAP - \cos A \sin \angle JAP + \sin \angle JAP} \\ &= \frac{(s-a)\sin A}{\sin A \cdot \dfrac{AP}{AJ} - \cos A \cdot \dfrac{PJ}{AJ} + \dfrac{PJ}{AJ}} \\ &= \frac{(s-a)\sin A}{AP \sin A + PJ(1-\cos A)} \cdot AJ. \end{aligned}$$

Thus, it suffices to prove $\dfrac{(s-a)\sin A}{AP\sin A + PJ(1-\cos A)} = \dfrac{1}{2}$. As J is the incentre of $\triangle BCD$, the lengths AP and PJ can be found easily.

Indeed, we first find that

$$CD = b - AD = b - 2c\cos A = b - \frac{b^2 + c^2 - a^2}{b} = \frac{a^2 - c^2}{b}.$$

Next, we have

$$CP = \frac{BC + CD - DB}{2} = \frac{a + \dfrac{a^2 - c^2}{b} - c}{2} = \frac{(a-c)s}{b}$$

and

$$AP = b - \frac{(a-c)s}{b}.$$

As PJ is the inradius of $\triangle BCD$, we have

$$\begin{aligned}
PJ &= \frac{2}{BC + CD + DB}[BCD] \\
&= \frac{1}{BC + CD + DB} \cdot BC \cdot CD \sin C \\
&= \frac{a \cdot \dfrac{a^2 - c^2}{b}}{a + \dfrac{a^2 - c^2}{b} + c} \cdot \sin C \\
&= \frac{a(a-c)(a+c)}{2(a+c)(s-c)} \cdot \frac{2[ABC]}{ab} \\
&= \frac{(a-c)[ABC]}{b(s-c)}.
\end{aligned}$$

Therefore,

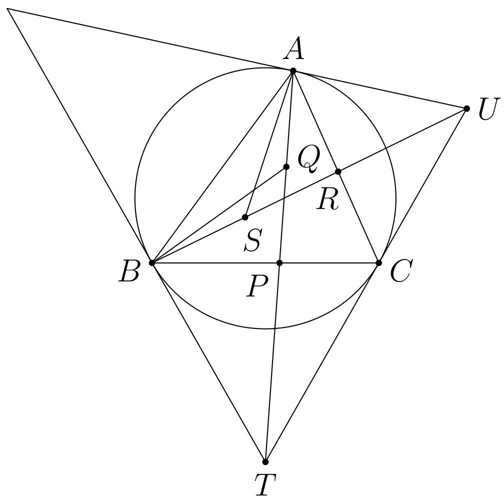
$$\begin{aligned}
& & \frac{(s-a)\sin A}{AP\sin A + PJ(1-\cos A)} &= \frac{1}{2} \\
&\Leftrightarrow & \left(b - \frac{(a-c)s}{b}\right)\sin A + \frac{(a-c)[ABC]}{b(s-c)} \cdot (1 - \cos A) - 2(s-a)\sin A &= 0 \\
&\Leftrightarrow & \left(b - \frac{(a-c)s}{b} - 2(s-a)\right)\sin A + \frac{(a-c)[ABC]}{b(s-c)} \cdot \left(1 - \frac{b^2 + c^2 - a^2}{2bc}\right) &= 0 \\
&\Leftrightarrow & \left(b - \frac{(a-c)s}{b} - 2(s-a)\right) \cdot \frac{2[ABC]}{bc} + \frac{(a-c)[ABC]}{b(s-c)} \cdot \frac{a^2 - (b-c)^2}{2bc} &= 0 \\
&\Leftrightarrow & \left(a - c - \frac{(a-c)s}{b}\right) \cdot \frac{2[ABC]}{bc} + \frac{(a-c)[ABC]}{b(s-c)} \cdot \frac{2(s-b)(s-c)}{bc} &= 0 \\
&\Leftrightarrow & -\frac{2(a-c)(s-b)[ABC]}{b^2c} + \frac{2(a-c)(s-b)[ABC]}{b^2c} &= 0.
\end{aligned}$$

This is obviously true. Hence, M is the midpoint of AJ, which means KL bisects the segment AJ. $\square$

Problem 4.5. (IMO Shortlist 2000 G5) The tangents at B and A to the circumcircle of an acute-angled triangle ABC meet the tangent at C at T and U respectively. AT meets BC at P, and Q is the midpoint of AP; BU meets CA at R, and S is the midpoint of BR. Prove that $\angle ABQ = \angle BAS$. Determine, in terms of ratios of side lengths, the triangles for which this angle is a maximum.

Analysis. For the equal angles, as the settings are the same, we just have to express $\angle BAS$ in terms of the side lengths of $\triangle ABC$. Note that the constructions of the points follow the order U, R, S. But when we compute $\angle BAS$, we should do it in the opposite order. In other words, we first convert everything that involves S to those which do not, so that the point S can be ignored. Then we do the same thing for the point R, etc. In this way, we can simplify the expressions step by step, until everything is expressed in terms of the information of $\triangle ABC$.

Solution.



By the ratio lemma, we have

$$\frac{AB \sin \angle BAS}{AR \sin \angle RAS} = \frac{BS}{RS} = 1.$$

This gives us an expression for $\angle BAS$. Indeed,

$$\begin{aligned}
& & AB \sin \angle BAS &= AR \sin \angle RAS \\
&\Rightarrow & c \sin \angle BAS &= AR \sin (A - \angle BAS) \\
&\Rightarrow & c \sin \angle BAS &= AR(\sin A \cos \angle BAS - \cos A \sin \angle BAS) \\
&\Rightarrow & (c + AR \cos A) \sin \angle BAS &= AR \sin A \cos \angle BAS \\
&\Rightarrow & \tan \angle BAS &= \frac{AR \sin A}{c + AR \cos A}. \qquad (1)
\end{aligned}$$

This gives an expression for $\tan \angle BAS$ that does not involve S. We may discard point S from our figure.

The next step is to find AR. One way is to apply the subtended angle theorem to $\triangle BAU$ since R lies on the segment BU. This gives

$$\frac{\sin\angle BAR}{AU} + \frac{\sin\angle UAR}{AB} = \frac{\sin\angle BAU}{AR}.$$

The angles can all be expressed using A, B, C. Indeed, this implies

$$\frac{\sin A}{AU} + \frac{\sin B}{c} = \frac{\sin(A+B)}{AR}.$$

Hence,

$$AR = \frac{\sin C}{\dfrac{\sin A}{AU} + \dfrac{\sin B}{c}} = \frac{cAU\sin C}{c\sin A + AU\sin B}. \tag{2}$$

Then we do not need point R anymore.

In isosceles $\triangle UAC$, we have $2AU\cos B = AC$, so that

$$AU = \frac{b}{2\cos B}. \tag{3}$$

The only task left is the calculation. Combining (1), (2) and (3), we have

$$\begin{aligned}
\tan\angle BAS &= \frac{AR\sin A}{c + AR\cos A} \\
&= \frac{\dfrac{cAU\sin C}{c\sin A + AU\sin B}\sin A}{c + \dfrac{cAU\sin C}{c\sin A + AU\sin B}\cos A} \\
&= \frac{AU\sin A\sin C}{c\sin A + AU\sin B + AU\cos A\sin C} \\
&= \frac{\dfrac{b}{2\cos B}\sin A\sin C}{c\sin A + \dfrac{b}{2\cos B}\sin B + \dfrac{b}{2\cos B}\cos A\sin C} \\
&= \frac{b\sin A\sin C}{2c\sin A\cos B + b\sin B + b\cos A\sin C}.
\end{aligned}$$

Each trigonometric function can be converted to a, b, c in the standard way. Indeed, this yields

$$\begin{aligned}
\tan\angle BAS &= \frac{b\cdot\dfrac{2[ABC]}{bc}\cdot\dfrac{2[ABC]}{ab}}{2c\cdot\dfrac{2[ABC]}{bc}\cdot\dfrac{a^2+c^2-b^2}{2ac} + b\cdot\dfrac{2[ABC]}{ac} + b\cdot\dfrac{b^2+c^2-a^2}{2bc}\cdot\dfrac{2[ABC]}{ab}} \\
&= \frac{4[ABC]}{2(a^2+c^2-b^2) + 2b^2 + (b^2+c^2-a^2)} \\
&= \frac{4[ABC]}{a^2+b^2+3c^2}.
\end{aligned}$$

As this expression is symmetric in a, b, we immediately get $\tan\angle BAS = \tan\angle ABQ$. Since the tangent function is injective on $(0°, 90°)$, this means $\angle BAS = \angle ABQ$. This establishes the first part.

To maximize $\angle BAS$, we have to maximize $\tan\angle BAS$ since $\angle BAS$ is acute and the tangent function is increasing on $(0°, 90°)$. Suppose $\dfrac{4[ABC]}{a^2+b^2+3c^2} \leqslant k$ for some $k > 0$. We first guess the value of k. WLOG, assume $c = 1$. As the expression is symmetric in a, b, probably we should consider the case $a = b$. Then $\triangle ABC$ is isosceles with

$$[ABC] = \frac{1}{2}\cdot 1\cdot\sqrt{a^2-\frac{1}{4}} = \frac{1}{4}\sqrt{4a^2-1}.$$

Then

$$\begin{aligned}
& \frac{4[ABC]}{a^2+a^2+3} \leqslant k \\
\Leftrightarrow\quad & \sqrt{4a^2-1} \leqslant 2ka^2+3k \\
\Leftrightarrow\quad & 4a^2-1 \leqslant 4k^2a^4+12k^2a^2+9k^2 \\
\Leftrightarrow\quad & (4k^2)a^4+(12k^2-4)a^2+(9k^2+1) \geqslant 0.
\end{aligned}$$

We would like the discriminant to be non-positive, i.e.

$$\Delta = (12k^2-4)^2-4(4k^2)(9k^2+1) \leqslant 0.$$

This is the same as $112k^2 \geqslant 16$, or $k \geqslant \dfrac{1}{\sqrt{7}}$. When $k = \dfrac{1}{\sqrt{7}}$, equality holds when $a = \sqrt{2}$. Therefore, we are going to show

$$\frac{4[ABC]}{a^2+b^2+3c^2} \leqslant \frac{1}{\sqrt{7}}$$

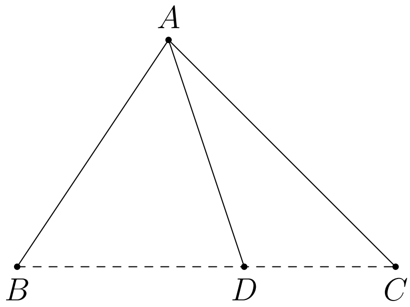
in general. Indeed,

$$\begin{aligned}
& \frac{4[ABC]}{a^2+b^2+3c^2} \leqslant \frac{1}{\sqrt{7}} \\
\Leftrightarrow\quad & \sqrt{-a^4-b^4-c^4+2a^2b^2+2b^2c^2+2c^2a^2} \leqslant \frac{1}{\sqrt{7}}(a^2+b^2+3c^2) \\
\Leftrightarrow\quad & 7(-a^4-b^4-c^4+2a^2b^2+2b^2c^2+2c^2a^2) \leqslant a^4+b^4+9c^4+2a^2b^2+6b^2c^2+6c^2a^2 \\
\Leftrightarrow\quad & 8a^4+8b^4+16c^4-12a^2b^2-8b^2c^2-8c^2a^2 \geqslant 0 \\
\Leftrightarrow\quad & 6(a^2-b^2)^2+2(a^2-2c^2)^2+2(b^2-2c^2)^2 \geqslant 0.
\end{aligned}$$

The step for completing squares is straightforward from the above equality case. This inequality is obviously true. The angle is maximized when $a : b : c = \sqrt{2} : \sqrt{2} : 1$. $\square$

Theorem 4.4. (Stewart's theorem(斯德瓦特定理)) Let D be a point in the interior of $\angle BAC$. Then D lies on the segment BC iff

$$AC^2 \cdot BD + AB^2 \cdot CD = (BD + CD)(AD^2 + BD \cdot CD).$$



Proof. Indeed, D lies on the segment BC iff $\angle ADB + \angle ADC = 180°$. This is equivalent to $\cos \angle ADB + \cos \angle ADC = 0$. By the cosine formula, this is the same as

$$\frac{AD^2 + BD^2 - AB^2}{2AD \cdot BD} + \frac{AD^2 + CD^2 - AC^2}{2AD \cdot CD} = 0.$$

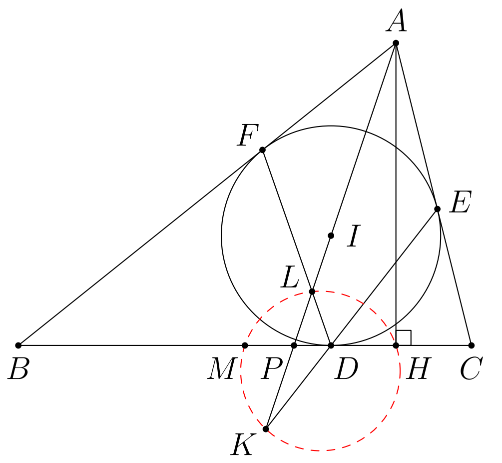
This can be simplified to give the desired equivalence. □

Stewart's theorem gives another way to prove the collinearity, but it is more useful to use the converse to find the length of a cevian. The equation in this theorem involves only side lengths. In practice, this usually gives tedious expressions since the equation has degree 3. However, it is still useful if there is no simple information about the angles.

Problem 4.6. (CWMO 2011 Problem 7) In triangle ABC with $AB > AC$ and incentre I, the incircle touches BC, CA, AB at D, E, F respectively. M is the midpoint of BC, and the altitude at A meets BC at H. Ray AI meets lines DE and DF at K and L, respectively. Prove that the points M, L, H, K are concyclic.

Analysis. To prove that four points are concyclic in an algebraic means, the simplest way is to use power. We let P be the intersection of MH and LK. Then we need to prove $PM \cdot PH = PL \cdot PK$. Note that each length can be found in terms of a, b, c. The idea of the following solution is rather straightforward. We also give an application of Stewart's theorem to find the length of an angle bisector.

Solution. Let AI meet BC at P. We first find the position of P on BC. By the angle bisector theorem, we have $\dfrac{BP}{CP} = \dfrac{c}{b}$. This gives $BP = \dfrac{ca}{b+c}$. From this, we can easily find PM and PH. Indeed, note that $BM = \dfrac{a}{2}$ and $BH = c\cos B = \dfrac{a^2 + c^2 - b^2}{2a}$.



This yields

$$PM = BP - BM = \frac{ca}{b+c} - \frac{a}{2} = \frac{a(c-b)}{2(b+c)} \tag{1}$$

and

$$PH = BH - BP = \frac{a^2+c^2-b^2}{2a} - \frac{ca}{b+c} = \frac{(c-b)\big((b+c)^2-a^2\big)}{2a(b+c)}. \tag{2}$$

Next, to find PL, it seems that using Menelaus' theorem is the simplest from the definition of L. We apply the theorem to $\triangle ABP$ and the line FLD. Using the usual lengths, this gives $\dfrac{AF}{FB} \times \dfrac{BD}{DP} \times \dfrac{PL}{LA} = 1$, i.e.

$$\frac{s-a}{s-b} \cdot \frac{s-b}{s-b-\dfrac{ca}{b+c}} \cdot \frac{PL}{LA} = 1.$$

Note that

$$s-b-\frac{ca}{b+c} = \frac{(c-b)(b+c-a)}{2(b+c)} = \frac{(c-b)(s-a)}{b+c}.$$

Therefore, we get $\dfrac{PL}{LA} = \dfrac{c-b}{b+c}$. We want to find an expression for PL which does not involve L. Therefore, we rewrite $LA = PA - PL$ and deduce $\dfrac{PL}{PA} = \dfrac{c-b}{2c}$. It follows that

$$PL = \frac{c-b}{2c} \cdot PA. \tag{3}$$

By symmetry,

$$PK = \frac{c-b}{2b} \cdot PA. \tag{4}$$

Note that M, L, H, K are concyclic iff $PM \cdot PH = PL \cdot PK$. From (1), (2), (3) and (4), we need to show

$$\frac{a(c-b)}{2(b+c)} \cdot \frac{(c-b)\big((b+c)^2-a^2\big)}{2a(b+c)} = \frac{c-b}{2c} \cdot PA \cdot \frac{c-b}{2b} \cdot PA.$$

After simplification, this is equivalent to

$$PA^2 = \frac{bc\big((b+c)^2 - a^2\big)}{(b+c)^2}.$$

This is a simple task since PA is simply the length of the angle bisector from A. Here, we try to use Stewart's theorem as an illustration. Since P lies on BC, we have

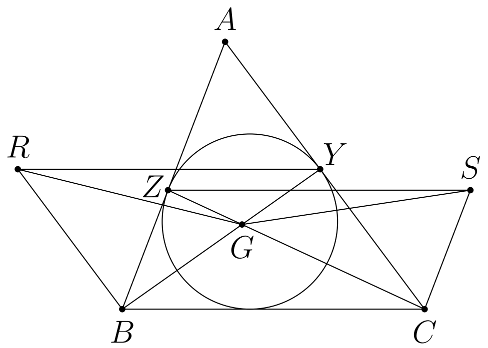
$$b^2 BP + c^2 CP = a(AP^2 + BP \cdot CP).$$

Using $BP = \dfrac{ca}{b+c}$ and $CP = \dfrac{ba}{b+c}$, we simplify and get $PA^2 = \dfrac{bc\big((b+c)^2 - a^2\big)}{(b+c)^2}$ as desired. $\square$

Problem 4.7. (IMO Shortlist 2009 G3) Let ABC be a triangle. The incircle of ABC touches the sides AB and AC at the points Z and Y, respectively. Let G be the point where the lines BY and CZ meet, and let R and S be points such that the two quadrilaterals $BCYR$ and $BCSZ$ are parallelograms. Prove that $GR = GS$.

Analysis. Again, we may focus on GS. No matter we consider $\triangle GCS$ or $\triangle GZS$, it seems that only the cosine formula is applicable.

Solution.



In $\triangle GZS$, by the cosine formula, we get

$$\begin{aligned} GS^2 &= GZ^2 + SZ^2 - 2GZ \cdot SZ \cos \angle GZS \\ &= GZ^2 + BC^2 - 2GZ \cdot BC \cos \angle BCZ \\ &= GZ^2 + a^2 - 2a \cdot GZ \cdot \frac{a^2 + CZ^2 - BZ^2}{2a \cdot CZ} \\ &= GZ^2 + a^2 - GZ \cdot \frac{a^2 + CZ^2 - BZ^2}{CZ}. \end{aligned} \tag{1}$$

Instead of using a, b, c, we shall convert all side lengths to $s-a$, $s-b$, $s-c$ since it would be more efficient in this case. Let $AY = AZ = s - a = x$, $BZ = s - b = y$

and $CY = s - c = z$. The length of the cevian CZ can be computed using Stewart's theorem. Applying this theorem to $\triangle ABC$ and the line AZB, we obtain

$$CZ^2 \cdot AB = CA^2 \cdot ZB + CB^2 \cdot ZA - AZ \cdot ZB \cdot AB.$$

This yields

$$CZ^2 = \frac{(z+x)^2 y + (y+z)^2 x - xy(x+y)}{x+y} = z^2 + \frac{4xyz}{x+y}.$$

Next, to compute GZ, we may find out the ratio that G cuts the segment CZ. For example, we apply Menelaus' theorem to $\triangle AZC$ and the line BGY. This yields $\frac{ZG}{GC} \times \frac{CY}{YA} \times \frac{AB}{BZ} = 1$, and so

$$\frac{ZG}{GC} = \frac{x}{z} \times \frac{y}{x+y} = \frac{xy}{yz+zx}.$$

This can be rewritten as

$$\frac{GZ}{CZ} = \frac{xy}{xy+yz+zx}. \tag{2}$$

From (1) and (2), we get

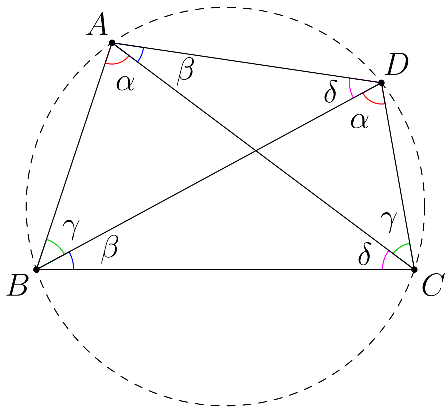
$$\begin{aligned}
&GS^2 \\
&= GZ^2 + a^2 - GZ \cdot \frac{a^2 + CZ^2 - BZ^2}{CZ} \\
&= \left(\frac{xy}{xy+yz+zx} \cdot CZ\right)^2 + (y+z)^2 - \frac{xy}{xy+yz+zx}\left((y+z)^2 + CZ^2 - y^2\right) \\
&= \left(\frac{x^2y^2}{(xy+yz+zx)^2} - \frac{xy}{xy+yz+zx}\right) CZ^2 + (y+z)^2 - \frac{xyz(2y+z)}{xy+yz+zx} \\
&= -\frac{xyz(x+y)}{(xy+yz+zx)^2} \cdot \left(z^2 + \frac{4xyz}{x+y}\right) + (y+z)^2 - \frac{xyz(2y+z)}{xy+yz+zx} \\
&= -\frac{xyz}{(xy+yz+zx)^2} \cdot (xz^2 + yz^2 + 4xyz) + (y+z)^2 - \frac{xy^2z}{xy+yz+zx} - \frac{xyz(y+z)}{xy+yz+zx} \\
&= -\frac{xyz}{(xy+yz+zx)^2} \cdot (xz^2 + yz^2 + 4xyz + xy^2 + y^2z + xyz) + (y+z)^2 - \frac{xyz(y+z)}{xy+yz+zx} \\
&= -\frac{xyz}{(xy+yz+zx)^2} \cdot \left(x(y^2+z^2) + yz(y+z) + 5xyz\right) + (y+z)^2 - \frac{xyz(y+z)}{xy+yz+zx}.
\end{aligned}$$

Since it is symmetric in y and z, we have $GR = GS$. □

Remarks. Note that we do not need to simplify the expression for GS^2, as we only need to look for an expression which is symmetric in y and z. This reduces a lot of calculation.

Theorem 4.5. (Ptolemy's theorem(托勒密定理)) Let $ABCD$ be a convex quadrilateral. Then A, B, C, D are concyclic iff

$$AB \cdot CD + BC \cdot DA = AC \cdot BD. \tag{1}$$



Proof. Suppose A, B, C, D are concyclic, lying on a circle with radius R. Denote $\angle BAC = \angle BDC = \alpha$, $\angle CAD = \angle CBD = \beta$, $\angle DBA = \angle DCA = \gamma$. Note that $\angle ADB = \angle ACB = \delta = 180° - (\alpha + \beta + \gamma)$. By the sine law, we obtain

$$\begin{aligned} AB \cdot CD + BC \cdot DA &= (2R\sin\delta)(2R\sin\beta) + (2R\sin\alpha)(2R\sin\gamma) \\ &= 4R^2\big(\sin(\alpha+\beta+\gamma)\sin\beta + \sin\alpha\sin\gamma\big) \end{aligned}$$

and

$$AC \cdot BD = \big(2R\sin(\beta+\gamma)\big)\big(2R\sin(\alpha+\beta)\big) = 4R^2\sin(\beta+\gamma)\sin(\alpha+\beta).$$

It suffices to show

$$\sin(\alpha+\beta+\gamma)\sin\beta + \sin\alpha\sin\gamma = \sin(\beta+\gamma)\sin(\alpha+\beta).$$

Indeed, we have

$$\begin{aligned} &\sin(\alpha+\beta+\gamma)\sin\beta + \sin\alpha\sin\gamma \\ =& -\frac{1}{2}\big(\cos(\alpha+2\beta+\gamma) - \cos(\alpha+\gamma)\big) - \frac{1}{2}\big(\cos(\alpha+\gamma) - \cos(\alpha-\gamma)\big) \\ =& -\frac{1}{2}\big(\cos(\alpha+2\beta+\gamma) - \cos(\alpha-\gamma)\big) \end{aligned}$$

and

$$\sin(\beta+\gamma)\sin(\alpha+\beta) = -\frac{1}{2}\big(\cos(\alpha+2\beta+\gamma) - \cos(\gamma-\alpha)\big).$$

Therefore, the two sides are equal and (1) is established.

The converse can be proved by inversion or complex numbers. We omit the proof here. □

Ptolemy's theorem is useful in relating the lengths of chords on a circle. Sometimes we can also use this result to prove the concyclicity if there is no information about the angles.

Problem 4.8. (CGMO 2016 Problem 2) In $\triangle ABC$, let $BC = a$, $AC = b$, $AB = c$ and Γ be its circumcircle.

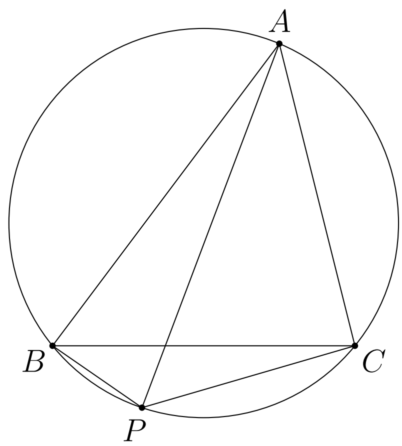
(1) Determine a necessary and sufficient condition on a, b and c, if there exists a unique point P ($P \neq B$, $P \neq C$) in the arc BC of Γ not passing through the point A, such that $PA = PB + PC$.

(2) Let P be the unique point described in (1). If AP bisects BC, prove that $\angle BAC < 60°$.

Analysis. Since the figure involves only four concyclic points, and the condition is related to lengths of chords, we may probably think of Ptolemy's theorem. This gives one more relation in the lengths PA, PB, PC.

Solution.

(1)

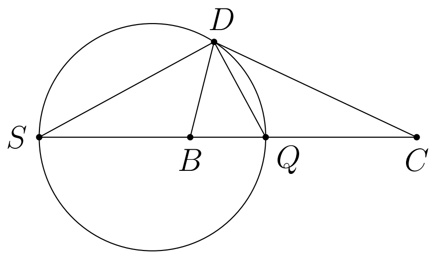


By Ptolemy's theorem, we have $aPA = bPB + cPC$. Then $PA = PB + PC$ iff $a(PB + PC) = bPB + cPC$. This is the same as

$$(a - b)PB + (a - c)PC = 0. \tag{1}$$

Firstly, in order that (1) has a solution, $a - b$ and $a - c$ must have different signs. WLOG, we may assume $b \geqslant a \geqslant c$. If one of $a - b$ and $a - c$ is 0, then so is the other, and hence $\triangle ABC$ is equilateral. In that case, P can be any point from (1). In particular, it is not unique. It suffices to consider $b > a > c$.

We claim that whenever $b > a > c$, there is a unique point P in the specified region satisfying (1). Indeed, for any such triangle, the ratio $\alpha = \dfrac{PB}{PC}$ is fixed. The locus of all points P with fixed $\dfrac{PB}{PC}$ is a circle ω whose centre lies on the line BC, which is known as an *Apollonius circle*(阿波羅尼斯圓).



To see this, note that there are two points Q and S on the line BC satisfying the condition $\frac{QB}{QC} = \frac{SB}{SC} = \alpha$ (say, Q lies on the segment BC and S lies outside BC). For any D not on the line BC such that $\frac{DB}{DC} = \alpha$, we know from the angle bisector theorem that DQ and DS are the angle bisectors (internal and external) of $\angle BDC$. Therefore, $\angle SDQ = 90°$ and D lies on the circle with diameter SQ.

Conversely, consider any point D lying on the circle with diameter SQ distinct from S, Q. By the ratio lemma, we have

$$\frac{DB\sin\angle BDQ}{DC\sin\angle CDQ} = \frac{QB}{QC} = \frac{SB}{SC} = \frac{DB\sin\angle BDS}{DC\sin\angle CDS}.$$

As $\angle SDQ = 90°$, this implies

$$\frac{\sin\angle BDQ}{\sin\angle CDQ} = \frac{\sin(90° - \angle BDQ)}{\sin(90° + \angle CDQ)},$$

i.e.

$$\frac{\sin\angle BDQ}{\sin\angle CDQ} = \frac{\cos\angle BDQ}{\cos\angle CDQ}.$$

Then $\tan\angle BDQ = \tan\angle CDQ$, and hence DQ bisects $\angle BDC$. This implies $\frac{DB}{DC} = \frac{QB}{QC} = \alpha$.

From the above fact, P must be an intersection point of ω and Γ. As ω cuts BC at two points, where exactly one lies on the segment BC, there are two intersection points of ω and Γ. Exactly one of them lies on $\overset{\frown}{BC}$ without A, which means P is uniquely determined.

Therefore, the necessary and sufficient condition for the statement to hold is $b > a > c$ or $c > a > b$. □

(2) Since AP bisects BC, we note from proposition 2.1 that $[ABP] = [ACP]$. This implies

$$\frac{1}{2}AB \cdot BP\sin\angle ABP = \frac{1}{2}AC \cdot CP\sin\angle ACP.$$

Since $\angle ABP + \angle ACP = 180^\circ$, we have $\sin \angle ABP = \sin \angle ACP$. Therefore, we get $cBP = bCP$. Together with (1), we find that $(a-b)b + (a-c)c = 0$. Thus,

$$a = \frac{b^2 + c^2}{b + c}.$$

To find some information about $\angle BAC$, we may use the cosine formula. Indeed,

$$\begin{aligned}
& \angle BAC < 60^\circ \\
\Leftrightarrow \quad & \cos \angle BAC > \frac{1}{2} \\
\Leftrightarrow \quad & \frac{b^2 + c^2 - a^2}{2bc} > \frac{1}{2} \\
\Leftrightarrow \quad & b^2 + c^2 - a^2 > bc \\
\Leftrightarrow \quad & b^2 + c^2 - \left(\frac{b^2 + c^2}{b + c}\right)^2 > bc \\
\Leftrightarrow \quad & 2bc(b^2 + c^2) > bc(b + c)^2 \\
\Leftrightarrow \quad & bc(b - c)^2 > 0.
\end{aligned}$$

This is true since we cannot have $b = c$ from the condition in (1). □

Problem 4.9. (IMO Shortlist 2010 G4/Problem 2) Let I be the incentre of triangle ABC and let Γ be its circumcircle. Let the line AI intersect Γ again at D. Let E be a point on the arc $\widehat{BDC}$ and F a point on the side BC such that

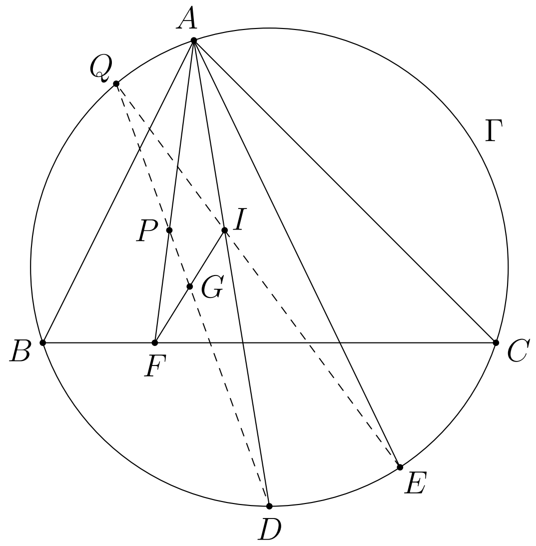
$$\angle BAF = \angle CAE < \frac{1}{2} \angle BAC.$$

Finally, let G be the midpoint of the segment IF. Prove that the lines DG and EI intersect on Γ.

Analysis. Let Q be the intersection of DG and EI. To show that Q lies on Γ, it seems to be the easiest to show Q, D, E, A are concyclic, as these points are more related from the figure. In particular, we can prove $\angle QDA = \angle QEA$. This only involves the lines DG, AD, IE, AE, and hence the exact position of Q is not important.

Let P be the intersection of DG and AF. Note that the pair of equal angles that we aim at proving is equivalent to the condition that $\triangle APD \sim \triangle AIE$. As we have $\angle PAD = \angle IAE$, it suffices to establish $\dfrac{AP}{AD} = \dfrac{AI}{AE}$. Hence, we only need to express each of these lengths in terms of a, b, c.

Solution.



Let P be the intersection of DG and AF, and let Q be the intersection of DG and EI. To find AP, from the definition of P, it is the easiest to apply Menelaus' theorem to $\triangle AFI$ and the line DGP. This implies

$$\frac{AP}{PF} \times \frac{FG}{GI} \times \frac{ID}{DA} = 1.$$

As we want to obtain an expression for AP which is not in terms of P, we change PF to $AF - AP$. After some manipulations, and by noting $FG = GI$, we get

$$AP = \frac{AD \cdot AF}{AD + ID}. \tag{1}$$

From (1) and our discussion, we need to find $AE \cdot AF$ and the lengths of AD, ID. Firstly, from $\angle ABF = \angle AEC$ and $\angle BAF = \angle EAC$, we have $\triangle ABF \sim \triangle AEC$. This yields

$$AE \cdot AF = bc. \tag{2}$$

(Again, one can obtain the same result using the sine law.)

Secondly, it is well-known that ID equals BD. By considering the isosceles $\triangle BCD$ where $\angle DBC = \dfrac{A}{2}$, we easily get

$$ID = BD = \frac{a}{2\cos\frac{A}{2}}. \tag{3}$$

Thirdly, the length AD can be found using Ptolemy's theorem, for example. Indeed, we have $aAD = bBD + cCD$. This implies

$$AD = \frac{b+c}{2\cos\frac{A}{2}}. \tag{4}$$

Combining (1), (2), (3) and (4), we have

$$\frac{AP}{AD}\cdot\frac{AE}{AI}=\frac{AE\cdot AF}{AI(AD+ID)}=\frac{bc}{(AD-ID)(AD+ID)}$$
$$=\frac{bc}{AD^2-ID^2}=\frac{4bc\cos^2\frac{A}{2}}{(b+c)^2-a^2}.$$

Note that

$$\cos^2\frac{A}{2}=\frac{\cos A+1}{2}=\frac{b^2+c^2-a^2+2bc}{4bc}=\frac{(b+c)^2-a^2}{4bc}.$$

Therefore, we get $\dfrac{AP}{AD}=\dfrac{AI}{AE}$. Together with

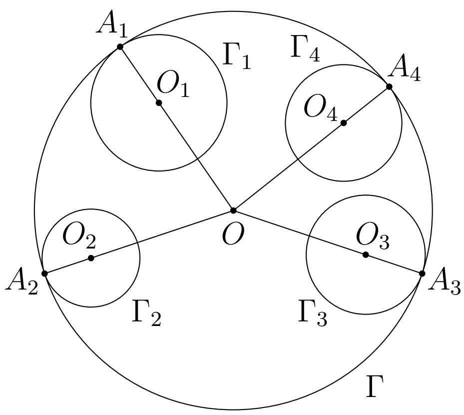
$$\angle PAD=\frac{A}{2}-\angle BAF=\frac{A}{2}-\angle CAE=\angle IAE,$$

we find that $\triangle PAD\sim\triangle IAE$. Therefore, $\angle PDA=\angle IEA$, and hence Q,D,E,A are concyclic. This shows Q lies on Γ as desired. $\square$

Remarks. Sometimes side length calculation and trigonometry can be used together with the usual tools in pure geometry like similar triangles. One can balance the amount of algebra and geometry in the steps depending on one's ability.

Theorem 4.6. (Casey's theorem(開世定理)) Given a circle Γ, let $\Gamma_1,\Gamma_2,\Gamma_3,\Gamma_4$ be smaller circles internally tangent to Γ in that order. For $1\leqslant i<j\leqslant 4$, let ℓ_{ij} be the length of an external common tangent of Γ_i and Γ_j. Then

$$\ell_{12}\ell_{34}+\ell_{23}\ell_{14}=\ell_{13}\ell_{24}. \tag{1}$$



Proof. Let O,O_1,O_2,O_3,O_4 and R,R_1,R_2,R_3,R_4 be the centres and radii of the circles $\Gamma,\Gamma_1,\Gamma_2,\Gamma_3,\Gamma_4$ respectively. For any $1\leqslant i<j\leqslant 4$, we have

$$\begin{aligned}\ell_{ij}^2&=O_iO_j^2-(R_i-R_j)^2\\&=O_iO_j^2-\big((R-OO_i)-(R-OO_j)\big)^2\\&=(OO_i^2+OO_j^2-2OO_i\cdot OO_j\cos\angle O_iOO_j)-(OO_i-OO_j)^2\\&=2OO_i\cdot OO_j(1-\cos\angle O_iOO_j).\end{aligned}$$

Next, we let A_1, A_2, A_3, A_4 be the contact points of $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ with Γ respectively. Then

$$A_iA_j^2 = R^2 + R^2 - 2R^2 \cos \angle A_iOA_j = 2R^2(1 - \cos \angle O_iOO_j).$$

Therefore, we get

$$\ell_{ij} = \frac{\sqrt{OO_i \cdot OO_j} \cdot A_iA_j}{R}.$$

Hence,

$$\begin{aligned}
\ell_{12}\ell_{34} + \ell_{23}\ell_{14} &= \frac{\sqrt{OO_1 \cdot OO_2} \cdot A_1A_2}{R} \cdot \frac{\sqrt{OO_3 \cdot OO_4} \cdot A_3A_4}{R} \\
&\quad + \frac{\sqrt{OO_2 \cdot OO_3} \cdot A_2A_3}{R} \cdot \frac{\sqrt{OO_1 \cdot OO_4} \cdot A_1A_4}{R} \\
&= \frac{\sqrt{OO_1 \cdot OO_2 \cdot OO_3 \cdot OO_4}}{R^2}(A_1A_2 \cdot A_3A_4 + A_2A_3 \cdot A_1A_4) \\
&= \frac{\sqrt{OO_1 \cdot OO_2 \cdot OO_3 \cdot OO_4}}{R^2} \cdot A_1A_3 \cdot A_2A_4 \\
&= \ell_{13}\ell_{24}.
\end{aligned}$$

Note that we have applied Ptolemy's theorem to the concyclic points A_1, A_2, A_3, A_4. □

In Casey's theorem, the circles Γ_j can be degenerated to a single point. When all of them degenerate to points, then this is simply Ptolemy's theorem. The most common application of Casey's theorem is the case when three of the circles are point circles.

There are many other configurations for Casey's theorem. Some of the circles Γ_j can be externally tangent to Γ instead of internally tangent. For any pair Γ_i, Γ_j, if both of them are externally/internally tangent to Γ, then we use the same ℓ_{ij} in (1). If one is externally tangent to Γ and the other is internally tangent to Γ, then ℓ_{ij} is defined to be the length of an internal common tangent of Γ_i and Γ_j.

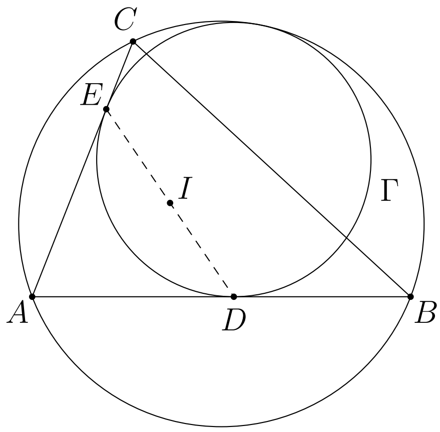
The converse of Casey's theorem is also true. That means if the circles Γ_j satisfy the equation (1), then there exists a circle Γ touching all of them.

Problem 4.10. Let I be the incentre of $\triangle ABC$. The unique circle Γ tangential to the sides AB, AC and the circumcircle of $\triangle ABC$ touches AB, AC at D, E respectively. Prove that D, E, I are collinear.

Analysis. The circle Γ is known as the A-*mixtilinear incircle*(偽內切圓) of $\triangle ABC$.

Since there is a pair of tangential circles, we may use Casey's theorem. For example, the circumcircle of $\triangle ABC$ can be regarded as tangential to the point circles A, B, C and Γ.

Solution.



For each point P outside Γ, let ℓ_P be the length of tangent from P to Γ. By Casey's theorem, we have

$$\ell_C \times AB + \ell_B \times AC = \ell_A \times BC.$$

This implies $cCE + bBD = ax$ where $AD = AE = x$. Note that

$$\begin{aligned} & & cCE + bBD &= ax \\ &\Rightarrow & c(b-x) + b(c-x) &= ax \\ &\Rightarrow & x &= \frac{2bc}{a+b+c}. \end{aligned}$$

To show that I lies on DE, as AI bisects $\angle DAE$ and $AD = AE$, it is the same as proving $AI = x\cos\frac{A}{2}$. Now,

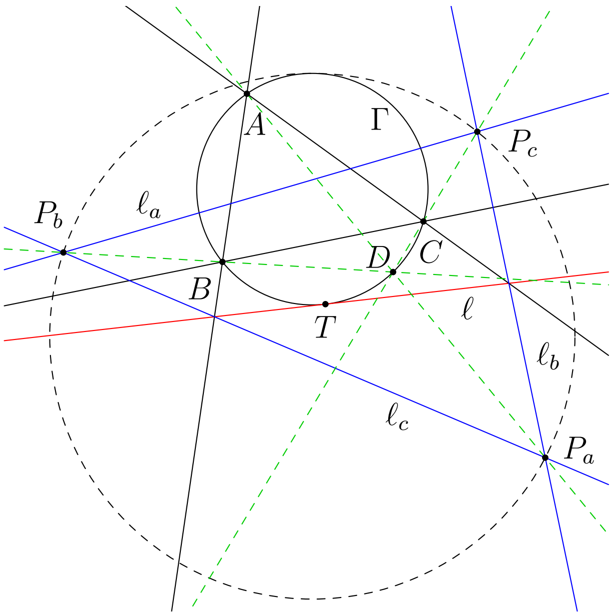
$$\begin{aligned} & & AI &= \frac{2bc}{a+b+c}\cos\frac{A}{2} \\ &\Leftrightarrow & \frac{r}{\sin\frac{A}{2}} &= \frac{2bc}{a+b+c}\cos\frac{A}{2} \\ &\Leftrightarrow & \frac{[ABC]}{s} &= \frac{2bc}{a+b+c}\sin\frac{A}{2}\cos\frac{A}{2} \\ &\Leftrightarrow & \frac{bc\sin A}{2s} &= \frac{bc}{a+b+c}\sin A. \end{aligned}$$

This is true, and hence D, E, I are collinear. $\square$

Problem 4.11. (IMO Shortlist 2011 G8/Problem 6) Let ABC be an acute triangle with circumcircle Γ. Let ℓ be a tangent line to Γ, and let ℓ_a, ℓ_b and ℓ_c be the lines obtained by reflecting ℓ in the lines BC, CA and AB, respectively. Show that the circumcircle of the triangle determined by the lines ℓ_a, ℓ_b and ℓ_c is tangent to the circle Γ.

Analysis. The assertion looks like the converse of Casey's theorem. We only need to compute the corresponding lengths. For the tangents, it is easier to compute the power from the vertices of the triangle determined by ℓ_a, ℓ_b, ℓ_c to Γ. This is a useful trick in using Casey's theorem since there are more ways to compute the power. For example, we can construct some nice secants from those points to the circles instead of using tangents.

Solution.



WLOG, assume ℓ touches Γ at T on $\widehat{BC}$ not containing A. Let P_c be the intersection of ℓ_a, ℓ_b, let P_a be the intersection of ℓ_b, ℓ_c, and let P_b be the intersection of ℓ_c, ℓ_a. Let t_x be the length of tangent from P_x to Γ for each $x = a, b, c$. We would like to show

$$t_b(P_cP_a) + t_c(P_aP_b) = t_a(P_bP_c). \tag{1}$$

Firstly, we can compute the interior angles of $\triangle P_aP_bP_c$. Indeed, we have

$$\begin{aligned}\measuredangle P_cP_aP_b &= \angle(\ell_b, \ell_c) = \angle(\ell_b, \ell) + \angle(\ell, \ell_c)\\ &= 2\angle(AC, \ell) + 2\angle(\ell, AB) = 2\angle(AC, AB) = 2\measuredangle CAB = -2\measuredangle BAC.\end{aligned}$$

As $\triangle ABC$ is acute, this shows $\angle P_cP_aP_b = 180^\circ - 2A$, and similarly for the other two angles. From this, we obtain

$$\frac{P_cP_a}{\sin(180^\circ - 2B)} = \frac{P_aP_b}{\sin(180^\circ - 2C)} = \frac{P_bP_c}{\sin(180^\circ - 2A)} = 2R'$$

where R' is the circumradius of $\triangle P_aP_bP_c$. Hence, (1) is equivalent to

$$t_b \sin 2B + t_c \sin 2C = t_a \sin 2A. \tag{2}$$

Secondly, as AB bisects ℓ and ℓ_c, AC bisects ℓ and ℓ_b, A is the excentre of the triangle determined by ℓ, ℓ_b, ℓ_c opposite to P_a. Hence, AP_a is the internal angle bisector of

$\angle P_cP_aP_b$. This holds true for other similar lines. Therefore, AP_a, BP_b, CP_c meet at the incentre of $\triangle P_aP_bP_c$. We call this point D. Also, we note that

$$\angle BDC = 90^\circ + \frac{1}{2}\angle P_bP_aP_c = 180^\circ - A.$$

This shows D lies on Γ. From this, we obtain $t_b = \sqrt{P_bB \cdot P_bD}$ since this is the square root of the power of P_b w.r.t. Γ.

Now, to compute P_bB, let d_b be the distance from B to ℓ_a. Note that it is also the distance from B to ℓ. We have

$$P_bB = \frac{d_b}{\sin\angle BP_bP_c} = \frac{d_b}{\sin(90^\circ - B)} = \frac{d_b}{\cos B}.$$

To compute P_bD, let r' be the inradius of $\triangle P_aP_bP_c$. Then

$$P_bD = \frac{r'}{\sin\angle BP_bP_c} = \frac{r'}{\cos B}.$$

From these, we find that

$$t_b = \frac{\sqrt{d_b \cdot r'}}{\cos B}. \tag{3}$$

Using (3) together with the other two similar equations, (2) holds iff

$$\frac{\sqrt{d_b}}{\cos B}\sin 2B + \frac{\sqrt{d_c}}{\cos C}\sin 2C = \frac{\sqrt{d_a}}{\cos A}\sin 2A.$$

This is equivalent to

$$\sin B\sqrt{d_b} + \sin C\sqrt{d_c} = \sin A\sqrt{d_a}.$$

It is not hard to prove this identity. Recall that d_b equals the distance from B to ℓ. By considering the corresponding right-angled triangle and using the fact that ℓ is tangent to Γ, we easily get $d_b = BT\sin\angle BAT = \frac{BT^2}{2R}$. Similar results hold for d_c and d_a. Note that

$$\begin{aligned}
& \sin B \cdot \frac{BT}{\sqrt{2R}} + \sin C \cdot \frac{CT}{\sqrt{2R}} = \sin A \cdot \frac{AT}{\sqrt{2R}} \\
\Leftrightarrow\quad & \frac{b}{2R}\cdot BT + \frac{c}{2R}\cdot CT = \frac{a}{2R}\cdot AT \\
\Leftrightarrow\quad & bBT + cCT = aAT.
\end{aligned}$$

The last equation is true by Ptolemy's theorem. Therefore, (1) holds from our steps, which shows the circumcircle of $\triangle P_aP_bP_c$ is tangent to Γ by the converse of Casey's theorem. □

Remarks. The same trick used in this solution can be applied to the following problem.

(IMO Shortlist 2020 G6) Let I and I_A be the incentre and the A-excentre of an acute-angled triangle ABC with $AB < AC$. Let the incircle meet BC at D. The line AD meets BI_A and CI_A at E and F, respectively. Prove that the circumcircles of triangles AID and I_AEF are tangent to each other.

Here summarizes the theorems introduced in this section.

- generalization of Pythagoras' theorem (problem 4.1)
- Carnot's theorem (problem 4.2)
- subtended angle theorem (problems 4.3, 4.4, 4.5)
- Stewart's theorem (problems 4.6, 4.7)
- Ptolemy's theorem (problems 4.8, 4.9, 4.11)
- Casey's theorem (problems 4.10, 4.11)

Links

Theorems

- Apollonius' theorem
- Carnot's theorem
- Casey's theorem
- Ceva's theorem
- Compound angle formulae
- Cosine formula
- Double angle formulae
- Euler's formula
- Generalization of Pythagoras' theorem
- Half angle formulae
- Heron's formula
- Iran lemma
- Menelaus' theorem
- Product-to-sum formulae
- Ptolemy's theorem
- Pythagoras' theorem
- Sine law
- Stewart's theorem
- Subtended angle theorem
- Sum-to-product formulae

Propositions

- 1.1: Properties of the sine function
- 1.2: Properties of the cosine function
- 1.3: Properties of the tangent function
- 1.4: Compound angle formulae
- 1.5: Double angle formulae
- 1.6: Half angle formulae
- 1.7: Product-to-sum formulae

- 1.8: Sum-to-product formulae
- 2.1: Ratio of areas
- 2.2: Area formulae for triangles
- 2.3: Ratio lemma
- 2.4: Properties of symmedians

<u>Problems</u>

- 1.1: $\sin 36°$
- 1.2: Compound angle formula for the tangent function
- 1.3: Identity in a regular heptagon
- 2.1: Omaforos Contest 2020 Problem 7
- 2.2: IMO Shortlist 2003 G4
- 2.3: IMO Shortlist 2008 G3
- 2.4: IMO Shortlist 2009 G7
- 2.5: Euler's formula
- 2.6: Application of Euler's formula
- 2.7: CSMO 2006 Problem 5
- 2.8: CGMO 2010 Problem 6(2)
- 2.9: CGMO 2012 Problem 2
- 2.10: Ratio lemma
- 2.11: Thailand TST 2014 Test 4 Problem 1
- 2.12: Injectivity of $\dfrac{\sin x}{\sin(\theta - x)}$
- 2.13: Isogonal lines
- 2.14: Sharygin Geometry Olympiad 2020 Correspondence Problem 11
- 3.1: Japan EGMO Preliminary 2015 Problem 2
- 3.2: IMO Prelim 2014 Problem 9
- 3.3: Sharygin Geometry Olympiad 2021 Final Grade 10-11 Problem 1
- 3.4: CMO 2005 Problem 2
- 3.5: Circle form of Ceva's theorem
- 3.6: Japan MO 2017 Problem 3
- 3.7: Iran lemma
- 3.8: IMO Shortlist 2017 G3

- 3.9: Proving collinearity using Ceva's theorem
- 4.1: China TST 2007 Quiz 1 Problem 2
- 4.2: IMO HK TST 2021 Test 2 Problem 3
- 4.3: Isogonal line lemma
- 4.4: IMO Shortlist 2006 G4
- 4.5: IMO Shortlist 2000 G5
- 4.6: CWMO 2011 Problem 7
- 4.7: IMO Shortlist 2009 G3
- 4.8: CGMO 2016 Problem 2
- 4.9: IMO 2010 Problem 2
- 4.10: Mixtilinear incircle
- 4.11: IMO 2011 Problem 6

Terminologies and Notations

- a, b, c
- r
- R
- s
- $[P_1P_2\cdots P_n]$
- Apollonius circle
- cosine function $\cos x$
- cyclocevian conjugate
- mixtilinear incircle
- sine function $\sin x$
- tangent function $\tan x$
- trigonometric functions

「此鳥不飛則已，一飛沖天；不鳴則已，一鳴驚人。」

《史記—滑稽列傳》